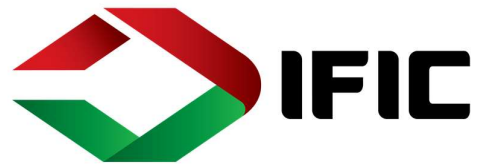


Information System Analysis and Design of “IFIC Bank PLC, Rajshahi Branch”



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Chapter 1

Recognition of Need

1.1 Introduction

IFIC Bank PLC is one of Bangladesh's prominent banking entities, initially established as a joint venture between the government and private sponsors in 1976. Originally intended as a finance company, it later transitioned into a full-fledged commercial bank in 1983, aligning with the government's decision to permit private sector banks. The Government of Bangladesh retains a significant stake in the bank, holding 32.75% of its share capital. Additionally, sponsors and directors, who bring extensive expertise in trade and commerce, own 4.11% of the share capital. The remaining shares are held by various institutions, both domestic and foreign, as well as general shareholders. Notably, the government is represented on the bank's board by three nominated directors. Together, sponsors and directors hold 36.86% of the total outstanding share capital.

The core values of IFIC Bank PLC emphasize integrity, fairness, innovation, and commitment. Integrity is upheld in all actions, ensuring honesty and transparency. The bank strives to serve its customers equitably with fairness, offering the best possible service. Innovation is encouraged, fostering a culture of creativity. The bank remains committed to excellence in customer service and the maximization of stakeholders' value, driven by a strong emphasis on teamwork.

The employee structure of IFIC Bank PLC, Rajshahi branch is given below:

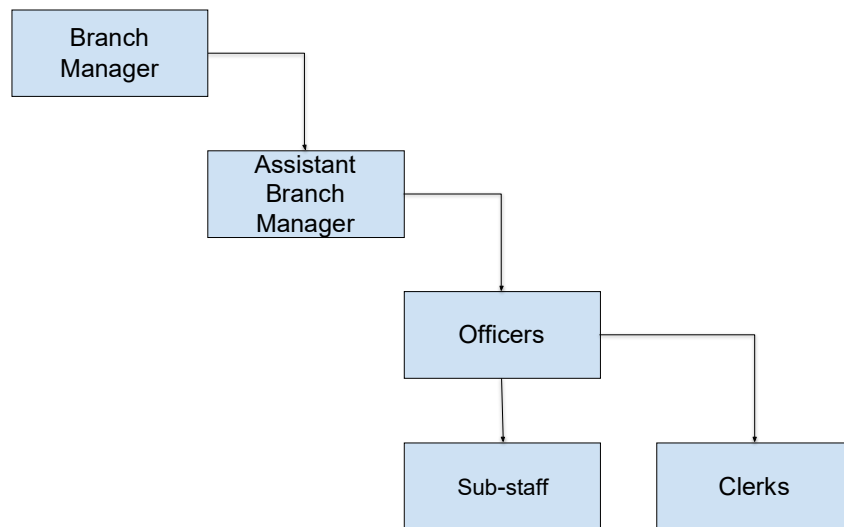


Figure 1.1: Employee Hierarchy of IFIC Bank PLC, Rajshahi Branch.¹

The vision of IFIC Bank PLC is to become the preferred financial service provider by fostering innovative, sustainable, and inclusive growth. The bank is committed to delivering best-in-class value to all stakeholders, ensuring excellence and progress in every aspect of its operations.

The mission of IFIC Bank Limited is to deliver exceptional service to clients through a skilled and dedicated workforce, whose creativity, innovation, and competitive edge set the bank apart in providing quality service to individuals and institutions. The bank is committed to the welfare and economic prosperity of the people and community, drawing inspiration from them for continued progress. In a competitive and complex financial environment, IFIC Bank focuses on ensuring growth and profitability for all stakeholders.

1.2 Problem Identification

To improve operational efficiency and customer satisfaction, it is essential to recognize the specific challenges faced by IFIC Bank PLC, Rajshahi Branch. Identifying these problems allows for targeted solutions that can enhance overall performance and service quality. Below

¹Image source: <https://www.ificbank.com.bd>

are key issues affecting the branch's operations and customer experience:

Problem 1: Inadequate Marketing and Customer Outreach

The Rajshahi Branch does not have a strong marketing strategy, limiting its visibility in the local market. Without targeted campaigns to reach potential customers, particularly in underserved areas, the branch struggles to expand its customer base. This lack of effective outreach not only hampers growth but also allows competitors to capture market share, further entrenching the branch's position in a highly competitive environment.

Problem 2: Lack of Credibility with Automated Fixed Deposit Receipt (FDR) System

The automated Fixed Deposit Receipt (FDR) system at IFIC Bank Ltd., Rajshahi Branch has faced several credibility issues. Customers have reported discrepancies between their expectations and the final FDR terms, leading to trust issues. Furthermore, a lack of transparency in the digital process and insufficient customer communication during FDR creation has made customers hesitant to fully adopt the automated system.

Problem 3: Inadequate Customer Background Checks in Loan Approval

IFIC Bank Ltd., Rajshahi Branch has occasionally approved loans without conducting thorough background checks on customers. This oversight increases the risk of loan defaults, as customers' financial histories and creditworthiness are not adequately assessed. Inadequate background checks can lead to financial losses for the bank and undermine its risk management practices, ultimately affecting its long-term sustainability.

Problem 4: Insufficient Manpower and Staffing Issues

Although IFIC Bank Ltd., Rajshahi Branch operates with a highly automated, software-based system, the lack of manpower affects customer support in helping customers adapt to these digital services. Many customers, particularly those unfamiliar with technology, struggle to navigate the automated systems without adequate guidance.

1.3 Conclusion

In this chapter, we recognized the operational challenges faced by IFIC Bank PLC, Rajshahi Branch, which are assessing its ability to meet customer expectations and achieve sustainable growth. Through the identification of key issues—such as inadequate marketing and cus-

tomer outreach, credibility concerns with the automated Fixed Deposit Receipt (FDR) system, inadequate customer background checks in the loan approval process, and insufficient manpower—these problems have highlighted critical areas that require immediate attention.

By addressing these challenges, the bank can improve customer trust, streamline processes, and achieve greater operational efficiency. These identified problems will form the foundation for deeper analysis in later chapters, where potential solutions and recommendations will be explored to enhance the overall performance of the branch and better align with the bank's long-term goals.

Chapter 2

Initial Feasibility Study

2.1 Introduction

An initial feasibility study involves evaluating a system proposal based on its organizational impact, workability, capacity to meet user needs, and efficient utilization of available resources. Following the initial investigation, the study is expanded to address three key factors: the system's problem, identification, and the evaluation of potential solutions. The Initial Feasibility Study for IFIC Bank PLC represents a pivotal phase in determining the feasibility of implementing a new system that aligns with the bank's operational requirements and strategic goals. This study provides a structured analysis to assess the proposed system's functionality, its potential effects on IFIC Bank's organizational framework, and its ability to satisfy both customer and internal user demands. Additionally, it examines the optimal use of resources in supporting the system's implementation. At this stage, the study focuses on evaluating the system's overall impact on the organization, its effectiveness in meeting user needs, and its resource efficiency. These aspects are discussed in detail in the subsequent sections.

2.2 Initial Feasibility Analysis

Before implementing any solutions, it is crucial to assess the feasibility of proposed changes. This analysis helps determine whether the bank's resources, technology, and operational capacity can support the necessary improvements. It provides a foundation for deciding the most practical and effective strategies moving forward.

2.2.1 Problem 1

2.2.1.1 Statement of the problem

Inadequate marketing and customer outreach have led to limited visibility and engagement in the local market.

2.2.1.2 Summary of findings

IFIC Bank PLC, Rajshahi Branch has struggled to implement effective marketing strategies and customer outreach programs. The lack of targeted campaigns and strong digital presence has limited its ability to attract new customers. This issue is particularly evident in underserved areas where awareness of the bank's services remains low.

2.2.1.3 Details of findings

The marketing and customer outreach efforts at IFIC Bank PLC, Rajshahi Branch are currently underperforming due to several factors. First, the branch lacks a focused marketing strategy that is aligned with modern banking trends and customer expectations. Its promotional efforts have largely been broad and untargeted, resulting in missed opportunities to engage specific customer segments, such as younger tech-savvy individuals, small businesses, and underserved communities. As a result, awareness of the bank's products and services remains low in certain key demographics.

The branch has not effectively utilized digital marketing tools, which are now a standard in the banking industry. In the era of social media and digital platforms, banks are expected to have a strong online presence, but IFIC Bank PLC has failed to fully capitalize on these channels. Social media platforms such as Facebook, Instagram, and LinkedIn are underutilized, meaning that the bank is missing out on reaching a wider audience, particularly young professionals who increasingly prefer online interactions over traditional banking methods. Similarly, the branch has not embraced email marketing or personalized messaging, which could help deliver tailored offers and services to customers based on their banking habits and preferences.

Furthermore, the branch lacks a robust data analytics system to track customer behaviors and tailor its offerings accordingly. Banks with effective marketing strategies often use customer data to design customized services and marketing campaigns. However, IFIC Bank PLC has not yet developed the infrastructure to collect and analyze such data, which hampers its ability

to provide personalized services. This lack of personalization has caused a disconnect between the bank and its customers, making it difficult to build strong, long-term relationships.

Moreover, there is little emphasis on community engagement and localized marketing. IFIC Bank PLC has not fully tapped into the potential of community-focused events or partnerships with local businesses, which could help increase its visibility and strengthen its connection with the local population. In a region like Rajshahi, where word-of-mouth and local trust are vital, the absence of such grassroots initiatives makes the bank less competitive compared to others that invest in community-building efforts.

Additionally, there is a noticeable absence of customer loyalty programs or incentive-based referral systems. Many banks have successfully implemented programs that reward loyal customers and encourage them to refer friends and family. IFIC Bank PLC could benefit from such programs to foster customer retention and attract new clients. Without these, the branch lacks a structured way to keep existing customers engaged and bring in new business organically.

Overall, the lack of a targeted marketing strategy, underutilization of digital platforms, insufficient use of data analytics, limited community engagement, and absence of loyalty programs have all contributed to the branch's poor performance in marketing and customer outreach. These gaps have allowed competitors to capture a larger share of the market, leaving IFIC Bank Ltd. struggling to grow its customer base.

2.2.1.4 Recommendation

To address this issue, IFIC Bank PLC, Rajshahi Branch should adopt a more data-driven and customer-centric marketing strategy. This includes leveraging data analytics to **create personalized offers, launching targeted campaigns for specific customer groups, and increasing the bank's digital marketing presence through social media and email campaigns**. The branch should also introduce **loyalty programs, referral incentives, and partnerships with local businesses** to enhance customer engagement and expand its market reach. These efforts can help improve brand visibility, attract new customers, and strengthen relationships with existing clients.

2.2.2 Problem 2

2.2.2.1 Statement of the problem

Lack of credibility with the automated FDR system has led to customer trust issues and discrepancies in FDR terms.

2.2.2.2 Summary of findings

IFIC Bank PLC, Rajshahi Branch has experienced credibility challenges with its automated Fixed Deposit Receipt (FDR) system. Customers have reported inconsistencies between their expectations and the final FDR terms, which has caused a decline in trust. The lack of transparency in the digital process and inadequate communication from the bank during the creation of FDRs has led to hesitation among customers in fully adopting the automated system.

2.2.2.3 Details of findings

Insufficient communication from the bank exacerbates the problem. Customers report that they receive minimal guidance during the digital FDR creation process, and often have to rely on bank staff for clarification after discrepancies arise. This delay in addressing issues further diminishes trust in the automated system, as customers feel they cannot rely on the process to deliver accurate and transparent results on its own.

A related issue is that customers, especially those less familiar with digital banking, find the automated system difficult to navigate. The bank has not provided sufficient educational resources or training to help users understand the FDR creation process. As a result, many customers are reluctant to adopt the system, preferring to stick with manual processes where they feel they have more control and visibility over their financial decisions. This reluctance limits the potential benefits of the automated system, as its full efficiency can only be realized if a majority of customers are willing to use it.

Moreover, the bank's response to these issues has been inadequate. While some errors in the FDR system are addressed manually after customers report them, there has been no systematic effort to improve the process or proactively resolve recurring issues. This reactive approach only serves to deepen customer dissatisfaction, as it implies that the bank is not fully committed to maintaining the credibility of its digital services.

In summary, the automated FDR system at IFIC Bank PLC, Rajshahi Branch is struggling with credibility due to discrepancies between expected and actual terms, lack of transparency in the digital process, and insufficient customer communication. These factors have created significant trust issues, leading many customers to avoid adopting the system altogether.

2.2.2.4 Recommendation

To restore trust in the automated FDR system, IFIC Bank PLC should take several steps. First, the bank must introduce **a more transparent process during FDR creation**, providing customers with a clear breakdown of terms, interest rates, and maturity dates before finalizing the transaction. Real-time verification systems can ensure that all details are correct and match the bank's advertised offers.

Second, communication with customers needs to be significantly improved. The bank should provide clear, step-by-step guidance during the FDR creation process, both through the system itself and via customer support. Additionally, **an educational campaign—using workshops, tutorials, and FAQs**—should be launched to help customers, especially those less familiar with technology, navigate the automated system with confidence.

Finally, customer support staff should be better trained to handle FDR-related queries and guide customers through the digital process, ensuring a seamless transition to the automated system and increasing customer confidence.

2.2.3 Problem 3

2.2.3.1 Statement of the problem

Inadequate customer background checks in loan approval have increased risk and financial losses.

2.2.3.2 Summary of findings

IFIC Bank PLC, Rajshahi Branch has issued loans without conducting thorough background checks, leading to an increased risk of defaults. Insufficient vetting of customers' financial histories and creditworthiness has resulted in financial losses and compromised the bank's risk management processes.

2.2.3.3 Details of findings

The loan approval process at IFIC Bank PLC, Rajshahi Branch has often bypassed critical background checks, such as comprehensive credit assessments and thorough evaluations of customers' financial histories. As a result, loans have been granted to individuals with questionable creditworthiness or insufficient financial stability. This has led to a significant rise in loan defaults, directly affecting the bank's profitability.

Moreover, the lack of robust verification mechanisms leaves the bank vulnerable to fraud and increases the overall financial risk. The branch's reliance on incomplete or outdated customer information further exacerbates the problem, as key indicators of a customer's ability to repay are overlooked. The absence of strict background checks compromises the branch's risk management framework and increases non-performing loans.

2.2.3.4 Recommendation

IFIC Bank PLC should strengthen its loan approval process by implementing **rigorous customer background checks, including credit scoring, income verification, and debt-to-income ratio analysis. Automated risk assessment tools** could also be introduced to enhance accuracy and efficiency. Additionally, the bank should ensure that updated financial records and credit reports are used before granting any loans, reducing the risk of defaults and safeguarding its financial stability.

2.2.4 Problem 4

2.2.4.1 Statement of the problem

Insufficient manpower and staffing issues have reduced customer support and adoption of digital services.

2.2.4.2 Summary of findings

Although IFIC Bank PLC, Rajshahi Branch has a highly automated system, the lack of sufficient manpower has impacted customer assistance. Customers unfamiliar with the digital platforms struggle to adapt, and there is a shortage of staff to guide them through the process, limiting the system's effectiveness and frustrating customers.

2.2.4.3 Details of findings

The branch operates with advanced digital systems, but many customers, particularly older individuals or those less familiar with technology, are unable to fully utilize these services without proper guidance. The limited number of staff available to help customers understand and navigate the digital platforms has resulted in underutilization of automated services.

Although the core banking operations remain efficient, the lack of customer support personnel creates bottlenecks when customers require help. Customers often face challenges in using services like online banking, automated deposit systems, and digital transactions without sufficient staff to assist. As a result, this gap in customer support slows down the branch's efforts to transition smoothly to a fully automated system, undermining the efficiency gains the digital infrastructure was intended to deliver.

Also people often feel that the reduced presence of bank staff and the emphasis on digital solutions compromise their confidence in the bank's ability to provide personalized, reliable service. This lack of human interaction can make customers feel less secure about the accuracy and security of their transactions, further deterring them from fully embracing the digital systems.

2.2.4.4 Recommendation

IFIC Bank PLC should invest in **increasing staffing levels specifically for customer support roles**, focusing on helping customers adapt to digital services. The bank could also introduce dedicated **"tech-support desks"** and offer digital literacy workshops to ensure customers are comfortable using the automated systems. Improved staff training on customer engagement would further enhance service delivery, ensuring both automation and human assistance are fully optimized.

2.3 Conclusion

In this chapter, we conducted a feasibility study to evaluate potential solutions for the key problems identified in the previous chapter: inadequate marketing and customer outreach, issues with the automated Fixed Deposit Receipt (FDR) system, inefficient loan approval processes, and insufficient manpower.

Our feasibility analysis has assessed each problem's economic, technical, and behavioural as-

pects. We have examined the potential benefits and costs associated with addressing these challenges, the technical requirements and constraints, and the likely impact on staff and customer acceptance. This preliminary analysis provides a foundation for understanding the feasibility of proposed solutions and highlights areas requiring further investigation.

By evaluating these feasibility considerations, we aim to develop targeted strategies that address the identified issues effectively. The insights gained from this analysis will guide the next steps in refining and implementing solutions that enhance operational efficiency, customer satisfaction, and overall branch performance.

Chapter 3

Analysis

3.1 Introduction

To analyze a system's feasibility, an analyst must gather detailed information using interviews, questionnaires, and on-site observations. These tools help assess the current system's performance and identify potential improvements for a new system.

Information collection involves stages: reviewing documents (e.g., procedure manuals), interviewing staff, and on-site observations. The analyst focuses on three key areas: the organization, the staff, and the workflow.

First, understanding the organization's policies, goals, and structure is essential. Policies define system rules, while the structure outlines management direction.

Second, the analyst examines the people operating the system, including their roles, relationships, and information needs, to facilitate effective communication.

Lastly, understanding the workflow is vital. This involves tracing data movement, often illustrated using a Data Flow Diagram (DFD).

In summary, a thorough analysis requires gathering information about the organization, staff, and workflow to provide useful system improvement recommendations.

3.2 Information gathering

The process of gathering information involves using specific tools known as information gathering tools. These tools are essential for analysts as they provide a structured approach to gather information accurately and comprehensively. They are carefully analyzed and followed to ensure effective information gathering.

3.2.1 Review of Literature, Procedure and Forms

Procedures, manuals, and forms serve as valuable sources for analysing an organization's system. Updated manuals can significantly streamline the information gathering process for analysts. During our analysis, we have gathered several manuals and forms that provide insights into the organization's functionality, publications, and collections. Since the organization we are examining is profit-oriented, many of the forms involve users. However, we have discovered a few specific forms and publications that have been instrumental in obtaining crucial information about the organization's working procedures, staff, and overall structure. Some of these relevant forms and manuals are included here for reference.

For ease of reference and further examination, we have included some of the most relevant forms and manuals. These materials not only provide context for understanding the procedural mechanisms within the organization but also serve as a foundation for future improvements and system optimizations.

Deposit Branch : _____

A/C with Branch : _____

Account Name : _____

Date : ____ / ____ / ____

Bank Copy

Account No. _____

(CD/SB/SND/FDR/MIS/SOD/CC)

Drawee Bank & Branch	Cheque Number & Date	Cash Denominations	Amount
		1000 x	
		500 x	
		200 x	
		100 x	
		50 x	
		20 x	
		10 x	
		5 x	
		2 x	
		1 x	
		Total Amount	

Depositor's Name: _____

Cell/Phone: _____

Depositor's Address: _____

Relation with A/C Holder: _____

Depositor's Signature _____

Receiving Officer _____

Authorized Officer _____

Note: (i) Please use separate Deposit Slip for cash & cheque; (ii) For clearing items use separate Deposit Slip.

In words: _____

Figure 3.1: Sample Money Receipt of Transaction.



ACCOUNT OPENING FORM (Individual)

For Bank Use Only															
Date	D	D	M	M	Y	Y	Y	Y							
Party ID															
Customer ID															
Account Number													-		

Manager, _____ Branch/Uposhakha

IFIC Bank PLC

Dear Sir/Madam,

I/we am/are applying to open an account in your Branch/ Uposhakha. All information required to Account Opening is given below:

Customer Photo

A. Account Related Information

Account Title (BLOCK LETTER)															
Account Title (Bengali)															
Type of Account	☐ Aamar ☐ Shohoj ☐ Savings ☐ Current ☐ FC ☐ Others (specify) _____														
Currency	☐ BDT ☐ USD ☐ GBP ☐ EUR ☐ JPY ☐ Nature of Account ☐ Individual ☐ Joint														
Initial Deposit (In Figure)											Initial Deposit (In Words)				
Other Facilities	☐ Debit Card ☐ SMS Alert ☐ IFIC Aamar Bank (Digital Banking) ☐ E-Statement Facility ☐ Cheque Book ☐ Number of Cheque Leaves: _____														

B. Personal Information

Mobile Number																E-mail (BLOCK LETTER)					
Account Holder's Name (BLOCK LETTER)																					
Account Holder's Name (Bengali)																					
Date of Birth	D	D	M	M	Y	Y	Y	Y	Gender	☐ Male ☐ Female ☐ Third Gender											
Father's Name																					
Mother's Name																					
Spouse's Name																					
Nationality (Passport copy with Visa must be obtained for foreign nationals)											Residential Status	☐ Resident ☐ Non-Resident (For Non-Resident Status, Guidelines for Foreign Exchange Transactions directives to be followed by the Bank)									
Monthly Income											Profession (Details)										
Source of Fund											TIN (if available)										
Present Address																					
Permanent Address																					
Contact/Mailing Address																					
Account Holder's Identity Document	National ID No.																				
	or																				
	☐ Passport ☐ Birth Certificate ☐ Driving License ☐ Employee ID ☐ Student ID																				
Introducer's Information (Applicable if account holder doesn't have NID)	Identity Document Number																				
	Introducer's Name																				
	Account No. of Introducer																				
	or																				
	NID Number of Introducer																				
	Introducer's Signature										Date										
	D	D	M	M	Y	Y	Y	Y													

1

Figure 3.2: Sample Copy of Personal Account Opening Form.

SMS Format		
Create Message > IFIC <Service Code> PIN > Send to 26969		
SMS User Guideline:		
Services	Hotkey/Service Code	SMS Example
Digital Services		
IFIC AB >>	Send 26969	
Account Services		
Account List	ACC	IFIC <SPACE> ACC
Balance Enquiry	BAL	IFIC<space>BAL
Mini Statement	STM	IFIC<space>STM
Foreign Exchange Rate Enquiry	CUR	IFIC<space>CUR
Fixed Deposit Rate Enquiry	FDR	IFIC<space>FDR
Interest Rate Enquiry		
For Savings Account	SBR	IFIC<space>SBR
For STD Account	STR	IFIC<space>STR
Card Services		
Debit Card	IFIC DC >>	Send 26969
Chequebook Request	IFIC CB >>	Send 26969
Note: For multiple A/C registered, put the sequence no. of the A/C as mentioned in the Registration Form after PIN and one space.		
Example: IFIC <space>BAL<space>PIN<space>2		
Available from all Telecom Operators.		

Figure 3.3: Several SMS Formats for Mobile Banking.¹Figure 3.4: Sample Advertisement Card.²¹Image source: <https://www.ificbank.com.bd>²Image source: <https://www.ificbank.com.bd>

Figure 3.5: Snapshot of Website.¹

In the course of our analysis, we have compiled a variety of manuals and forms that offer significant insight into the organization's internal operations, published materials, and data collections. Since the organization we are analyzing is profit-driven, many of the forms are inherently designed to engage and involve the end-users or customers in its processes. These user-oriented forms often provide critical information about how the organization interacts with its clientele, processes transactions, and manages service delivery.

However, beyond the user-centric forms, we have discovered several specific forms and publications that have proven instrumental in offering a deeper understanding of the organization's inner workings. These documents have been crucial in helping us analyze the organization's working procedures, the structure and roles of its staff, and its overall operational framework. By examining these forms, we can better comprehend how the bank's system functions, how information flows across departments, and how decisions are made at various levels.

3.2.2 On Site Observation

Another information-gathering tool used in system studies is on-site observation. It is the process of recognizing and noting people, objects, and occurrences to obtain information. The major objective of on-site observation is to get as close as possible to the "real" system being studied. For this reason, it is important that the analyst is knowledgeable about the general makeup and activities of the system.

¹Image source: <https://www.ificbank.com.bd>

Authority of IFIC Bank PLC, Rajshahi Branch

IFIC Bank PLC is a Bangladeshi private commercial bank headquartered in Dhaka. The bank provides a range of products and services including retail banking, corporate finance, SME banking, women banking, digital banking, asset management, equity brokerage, and security. The branch is managed by the Branch Manager, under whom the following key personnel operate:

- **CSM (Customer Service Manager)**
 - **ACSM (Associate Customer Service Manager)**
 - **CSO (Customer Service Officers):** Responsible for providing customer service, handling account opening, fund transfers, cheque issuance, cheque transfer, and card-related issues.
- **CSM (Customer Service Manager):** Involved in cash management and remittance activities.
- **BOM (Branch Operation Manager):** Oversees operational tasks, cash management, vault management, compliance, anti-money laundering measures, and the implementation of internal circulars.
- **RM (Relationship Manager):** Focuses on maintaining relationships with existing clients, acquiring new clients, selling banking products, booking new deposits, and mobilizing business.

Surroundings of IFIC Bank PLC, Rajshahi Branch

The surroundings of the IFIC Bank PLC, Rajshahi Branch, reflect a bustling yet organized environment. Situated in a prominent area of Rajshahi, the branch is easily accessible to both pedestrians and vehicles. The bank is surrounded by various commercial establishments, including shops, offices, and restaurants, contributing to the lively atmosphere of the neighborhood. Outside the branch, there are ATMs available for quick banking services, while security personnel and CCTV cameras ensure a safe and secure environment. The overall ambiance combines a professional banking experience with the vibrancy of the local community.



Figure 3.6: Entrance of the Bank.

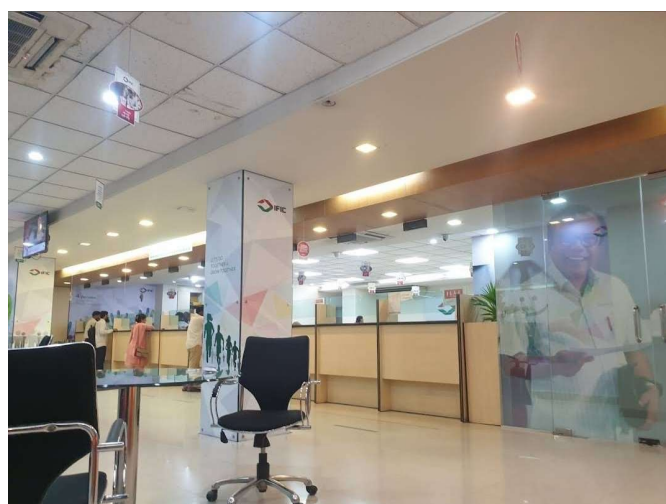


Figure 3.7: Inside the Office.

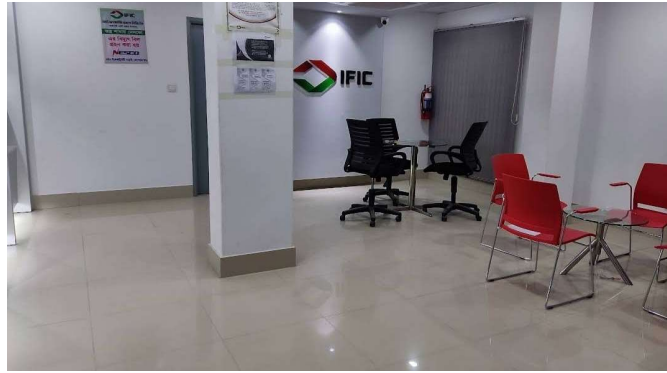


Figure 3.8: Waiting Area 1.



Figure 3.9: Waiting Area 2.



Figure 3.10: Waiting Area 3.

Direct Observation of IFIC Bank PLC, Rajshahi Branch

- The organization has a single, well-decorated entrance gate.
- A security guard table is positioned at the main building's entrance for inspection.
- The entire building is under CCTV surveillance.
- There appears to be a positive relationship between staff and system management.
- Office attendants and security guards display well-mannered behavior and are helpful and responsible in their duties.
- The employees are highly qualified.
- The overall environment is neat, clean, and well-maintained.

3.2.3 Interviews

The interview is done with Robiul Islam, Branch Manager, IFIC Bank PLC, Rajshahi Branch.

1. What core banking software does IFIC Bank use to manage daily operations?

Answer: IFIC Bank uses a proprietary core banking solution to manage all its daily operations.

2. How does the bank handle system updates and maintenance?

Answer: Updates are handled by the IT department, with regular patches and system

checks to ensure smooth functioning.

3. How is customer data stored and managed within the bank's systems?

Answer: Data is stored securely in centralized databases with restricted access and encryption.

4. Can you describe the level of automation in IFIC Bank's operations?

Answer: A significant portion of the operations is automated, especially in transaction processing and reporting.

5. How does the bank ensure system reliability and uptime?

Answer: The bank maintains high uptime with redundant systems and constant monitoring.

6. What measures are in place to protect customer data from cyber threats?

Answer: The bank uses multi-layered firewalls, encryption, and intrusion detection systems.

7. How does IFIC Bank monitor and prevent unauthorized access to its systems?

Answer: Access control is implemented using biometrics, strong passwords, and role-based permissions.

8. What protocols are followed in the event of a system breach or failure?

Answer: The bank follows a detailed incident response plan, involving immediate isolation, investigation, and reporting.

9. How does the bank ensure compliance with financial regulations on data security?

Answer: Compliance is ensured through regular audits and alignment with national and international data protection standards.

10. What kind of disaster recovery plan is in place for critical system failures?

Answer: There is an off-site backup system with predefined recovery procedures in case of critical failures.

11. How are digital transactions processed and verified at IFIC Bank?

Answer: Digital transactions are processed in real-time, verified using secure encryption protocols.

12. What mechanisms ensure the accuracy and speed of inter-bank transfers?

Answer: Mechanisms like SWIFT and RTGS ensure accurate and fast transfers.

13. How does the bank handle system downtime affecting transaction processing?

Answer: Backup servers take over during system downtime, ensuring minimal disruption in transactions.

14. What CRM system is used to track and manage customer interactions?

Answer: IFIC Bank uses an integrated CRM system to track customer interactions and manage relationships.

15. How does the bank's system help in providing personalized services to customers?

Answer: The system uses customer data analytics to offer tailored banking services.

16. How does IFIC Bank manage complaints or service requests through its system?

Answer: Complaints are logged into the CRM and routed to the appropriate department for resolution.

17. How secure is IFIC Bank's mobile banking platform?

Answer: The mobile platform uses two-factor authentication and encryption to secure transactions.

18. What kind of real-time services are offered via online banking?

Answer: Services like fund transfers, bill payments, and account monitoring are available in real-time.

19. How is customer feedback incorporated into improving the online banking system?

Answer: Feedback from customers is regularly analyzed to make system improvements and enhance user experience.

20. How do branch systems integrate with the central system of IFIC Bank?

Answer: Branch systems are fully integrated with the central system, ensuring seamless data flow.

21. What kind of system support do branch employees receive during high-volume transactions?

Answer: IT support is provided via a help desk, and real-time troubleshooting is available for branch employees.

22. How does the bank ensure the security of its branch systems?

Answer: All branch systems are secured with firewalls, regular audits, and employee training on security protocols.

23. Are there any upcoming technological improvements planned for the banking systems?

Answer: The bank is planning to upgrade its mobile banking platform and explore blockchain technology for transaction security.

24. How is artificial intelligence or machine learning being integrated into the bank's operations?

Answer: AI is being tested in customer support systems and fraud detection.

25. What challenges does IFIC Bank face in adapting to new financial technologies?

Answer: Adapting to new financial technologies is challenging due to regulatory constraints and the need for continuous staff training.

26. Can you explain the overall architecture of IFIC Bank's IT system and how different components (core banking, CRM, mobile banking, etc.) integrate with each other?

Answer: The bank uses a multi-layered system architecture where the core banking platform integrates with CRM, mobile banking, and other digital services through APIs. This ensures real-time data synchronization across different platforms, allowing seamless operations between branches, online banking, and other services.

27. How does the bank manage large volumes of customer and transaction data, and what strategies are in place to ensure data integrity and security across all systems?

Answer: IFIC Bank stores customer and transaction data in a centralized database with multiple backups. Data security measures include encryption, role-based access, and

regular audits. Encryption is applied both in transit and at rest, with strict protocols to protect sensitive data.

28. What digital transformation initiatives has IFIC Bank implemented recently, and how have these initiatives improved operational efficiency or customer satisfaction?

Answer: The bank has launched several digital transformation initiatives, including a mobile banking app, online account management tools, and automated teller machines (ATMs) with enhanced capabilities. These changes have streamlined operations and enhanced customer experience through more self-service options.

29. What are the key cybersecurity challenges faced by IFIC Bank, and how does the bank's system safeguard against new and emerging cyber threats?

Answer: Key cybersecurity challenges include protection against phishing attacks, ransomware, and insider threats. The bank employs a layered defense strategy, including firewalls, intrusion detection systems (IDS), multi-factor authentication (MFA), and constant monitoring for suspicious activities.

30. How does IFIC Bank ensure that its systems can scale to meet growing customer demand, particularly during peak times like year-end or special promotions?

Answer: To ensure scalability, the bank utilizes cloud services and load balancers to distribute traffic across multiple servers, particularly during peak periods. Continuous monitoring and timely system upgrades ensure consistent service and growing customer demand handling.

31. Can you describe the bank's disaster recovery plan and how it ensures continuity of operations in the event of a critical system failure?

Answer: The bank has a robust disaster recovery plan, including an off-site data center where critical systems are backed up. In case of a major outage, operations can be shifted to this backup location with minimal disruption. Regular drills and tests ensure readiness for unexpected events.

32. How does IFIC Bank ensure that its IT systems comply with local and international financial regulations, particularly regarding data privacy and anti-

money laundering (AML) protocols?

Answer: The bank adheres to local regulations such as the Bangladesh Bank's guidelines on data security and AML, as well as international standards like GDPR. Automated compliance checks and regular audits ensure ongoing adherence to these regulations.

33. How has IFIC Bank evolved its mobile and internet banking platforms to meet changing customer expectations, and what challenges have you faced during this evolution?

Answer: IFIC Bank has expanded its mobile and internet banking platforms to include services like instant fund transfers, bill payments, and account management. A key challenge is maintaining security while ensuring ease of use. Continuous customer feedback drives ongoing system improvements.

34. Is the bank currently utilizing artificial intelligence or machine learning in its operations? If so, how are these technologies being applied (e.g., fraud detection, customer support, etc.)?

Answer: AI and machine learning are used in fraud detection and customer service chatbots. Fraud detection algorithms identify suspicious transactions, while chatbots handle customer queries on the mobile banking platform, reducing human staff workload.

35. How often are system updates and maintenance performed, and what processes are in place to minimize downtime and service disruptions during these updates?

Answer: Maintenance and upgrades are scheduled during non-peak hours to minimize service disruption. The bank follows a regular maintenance cycle, updating systems with security patches and performance improvements. Customers are notified in advance of major upgrades to avoid inconvenience.

36. What role does the IT system play in improving customer experience at IFIC Bank, and how do you measure the success of these initiatives?

Answer: The IT system enhances customer experience by providing services like 24/7 access to banking, personalized notifications, and faster transaction processing. Success is measured through customer satisfaction surveys, feedback on digital platforms, and user engagement metrics.

37. How do you ensure that employees are well-trained to use the bank's systems efficiently? Are there any ongoing training programs or resources?

Answer: Employees undergo regular training on system updates and new features. E-learning modules, workshops, and hands-on training sessions are provided. Training programs are continuously updated to keep pace with technological advancements.

38. Does IFIC Bank collaborate with third-party IT vendors for system development or security? If so, how do you manage vendor relationships and ensure data protection?

Answer: The bank collaborates with third-party vendors for specialized services like cybersecurity, software development, and infrastructure management. Vendor relationships are managed through contracts with strict data protection clauses, and regular audits ensure compliance with security standards.

39. What future technological innovations or upgrades is IFIC Bank planning to implement to stay competitive in the banking sector?

Answer: IFIC Bank is exploring blockchain for secure transactions and advanced AI for personalized banking services. Upcoming upgrades include enhancing the mobile banking app and expanding digital wallet use to align with changing consumer behavior.

40. What are the major challenges the bank faces with its current systems, and what improvements or changes would you prioritize in the near future?

Answer: Challenges include integrating legacy systems with newer technologies while maintaining data security. Priority improvements include increasing system automation and enhancing mobile and online banking user interfaces.

3.2.4 Questionnaires

We have prepared a detailed questionnaire to capture essential feedback from both clients and staff. This survey aims to assess the effectiveness of the current systems and identify opportunities for improvement.

1. What is the most effective way to improve customer satisfaction?

Answer:_____.

2. What can be done to increase productivity?

Answer:_____.

3. What is the most convenient feature of the mobile banking app?

Answer:_____.

4. What is the biggest obstacle getting/giving the customer service

Answer:_____.

5. Name one feature that you would like to add to the current service.

Answer:_____.

6. Does IFIC Bank have a dedicated team for monitoring fraud prevention?

Answer: Yes / No

7. Is there a process in place for customers to report suspicious activities on their accounts?

Answer: Yes / No

8. Are regular updates implemented to enhance the bank's cybersecurity measures?

Answer: Yes / No

9. Does IFIC Bank provide educational resources on safe online banking practices for customers?

Answer: Yes / No

10. Is biometric authentication (e.g., fingerprint, facial recognition) available on the mobile banking app?

Answer: Yes / No

11. Rank the following priorities for IFIC Bank's system enhancements from 1 (most important) to 5 (least important).

- _____ Data security
- _____ Transaction speed
- _____ Customer experience
- _____ System uptime
- _____ Automation of operations

12. Rank these factors in terms of their importance for IFIC Bank's mobile banking app from 1 (most important) to 5 (least important).

- _____ User-friendliness
- _____ Security features
- _____ Transaction processing time
- _____ Range of available services
- _____ Customer support integration

13. Rank the following challenges IFIC Bank faces with its current system infrastructure from 1 (most critical) to 5 (least critical).

- _____ Legacy system integration
- _____ Data security concerns
- _____ High maintenance costs
- _____ Scalability issues
- _____ Employee training

14. Rank the importance of the following features for IFIC Bank's customer-facing digital platforms from 1 (most important) to 5 (least important).

- _____ Mobile app user interface
- _____ Speed of online transactions
- _____ Security of online transactions
- _____ Availability of customer support
- _____ Range of services offered (e.g., loan applications, fund transfers)

15. Rank the focus areas for IFIC Bank's future IT investments from 1 (top priority) to 5 (lowest priority).

- _____ Cloud infrastructure
- _____ AI-driven customer support

- _____ Data analytics and reporting tools
- _____ Enhancing cybersecurity
- _____ Upgrading mobile banking features

16. Which of the following is the primary focus of IFIC Bank's digital transformation?

- A) Enhancing mobile banking
- B) Expanding physical branches
- C) Introducing blockchain technology
- D) Streamlining internal operations

17. What is the most common method for customers to access their accounts at IFIC Bank?

- A) Mobile app
- B) Online banking website
- C) ATM
- D) In-branch visits

18. How often do IFIC Bank customers use digital banking services?

- A) Daily
- B) Weekly
- C) Monthly
- D) Rarely

19. Which feature do customers value most in IFIC Bank's mobile banking app?

- A) Instant fund transfers
- B) Budgeting tools
- C) Account balance notifications
- D) Customer support chat

20. What is the primary reason customers choose IFIC Bank over its competitors?

- A) Competitive interest rates
- B) Quality of customer service
- C) Range of services offered
- D) Accessibility of digital platforms

21. How would you rate the overall reliability of IFIC Bank's core banking system?

- ☐ 1 Poor
- ☐ 2 Fair
- ☐ 3 Good
- ☐ 4 Very Good
- ☐ 5 Excellent

22. Rate the effectiveness of IFIC Bank's cybersecurity measures.

- ☐ 1 Poor
- ☐ 2 Fair
- ☐ 3 Good
- ☐ 4 Very Good
- ☐ 5 Excellent

23. How would you rate the user experience of IFIC Bank's mobile banking app?

- ☐ 1 Poor
- ☐ 2 Fair
- ☐ 3 Good
- ☐ 4 Very Good
- ☐ 5 Excellent

24. Rate the speed and accuracy of digital transactions at IFIC Bank.

- ☐ 1 Poor
- ☐ 2 Fair
- ☐ 3 Good
- ☐ 4 Very Good
- ☐ 5 Excellent

25. How satisfied are you with the bank's system uptime during high-demand periods?

- ☐ 1 Poor
- ☐ 2 Fair
- ☐ 3 Good
- ☐ 4 Very Good
- ☐ 5 Excellent

3.3 Information Representation/Analysis

In this section, we present various charts and graphs that visually represent the data collected from our questionnaires, alongside Data Flow Diagrams (DFDs). These visualizations and diagrams facilitate the analysis and interpretation of responses, offering a clear and comprehensive overview of key findings and system interactions. The DFDs provide additional insights into the flow of information within the system, enhancing our understanding of operational processes and user interactions.

Furthermore, the graphical representations aim to highlight patterns and trends that are crucial for understanding user behavior and system efficiency. Bar charts, pie charts, and line graphs are used to depict quantitative data such as response rates, user satisfaction levels, and the frequency of system interactions. In addition, the DFDs serve as a tool to map out data movement and processing, which helps to identify potential bottlenecks and areas for optimization within the system. This combination of visual and analytical tools supports a deeper exploration of the system's functionality and user experience.

3.3.1 Data Visualization: Charts and Graphs from Questionnaire Responses

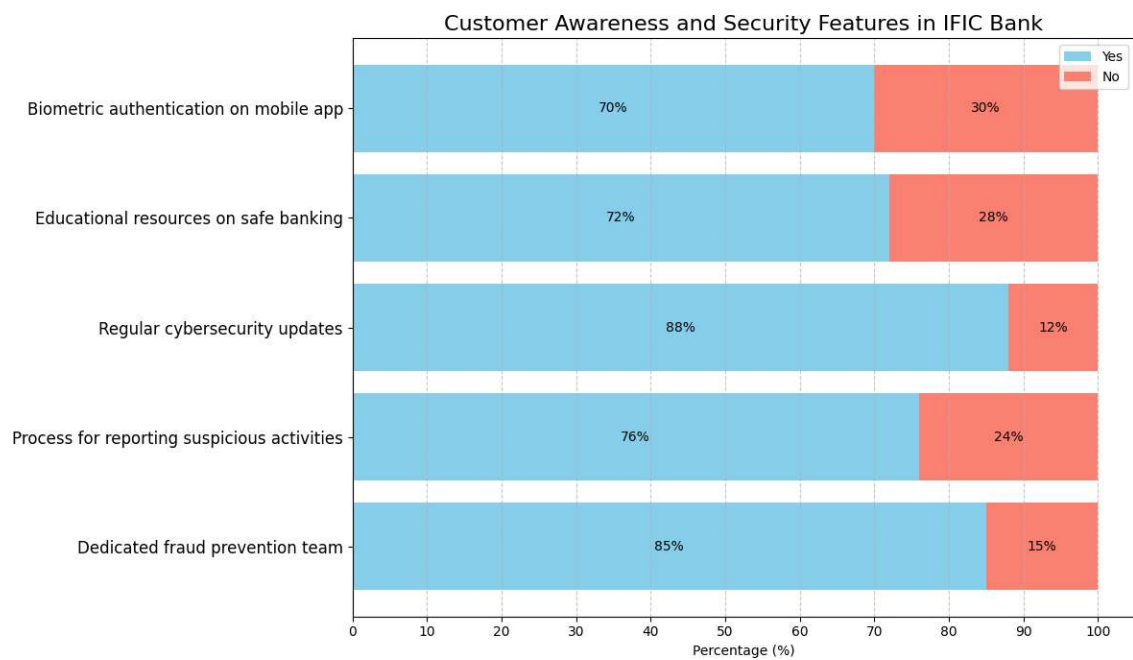


Figure 3.11: Distribution of Responses to Key Security and Customer Service Features.

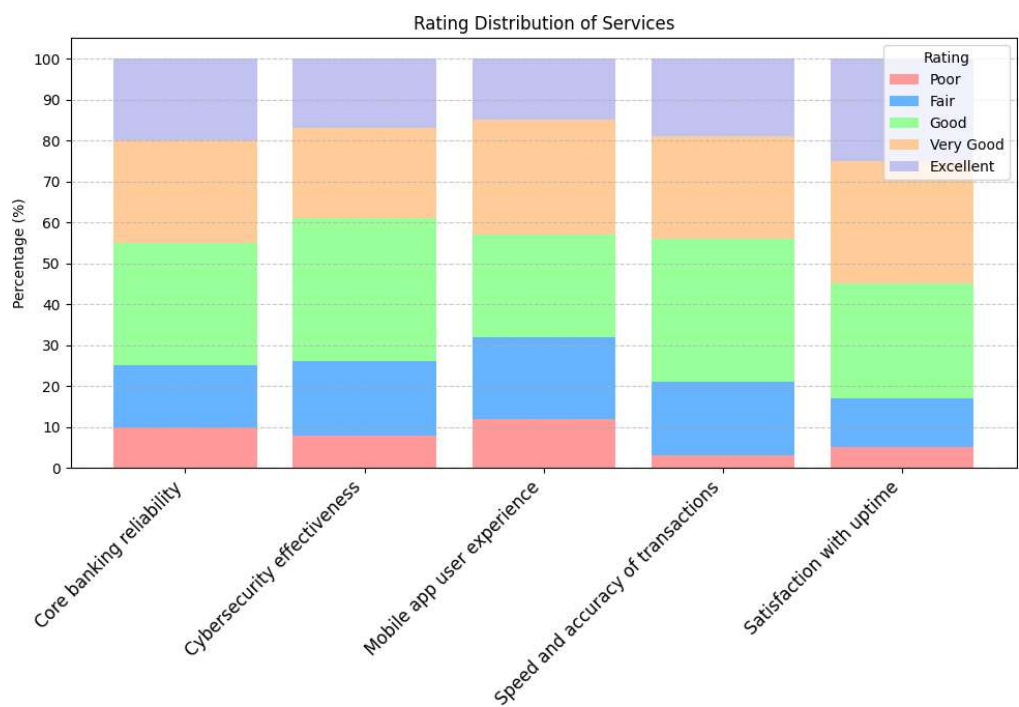


Figure 3.12: Ratings Distribution of Services.

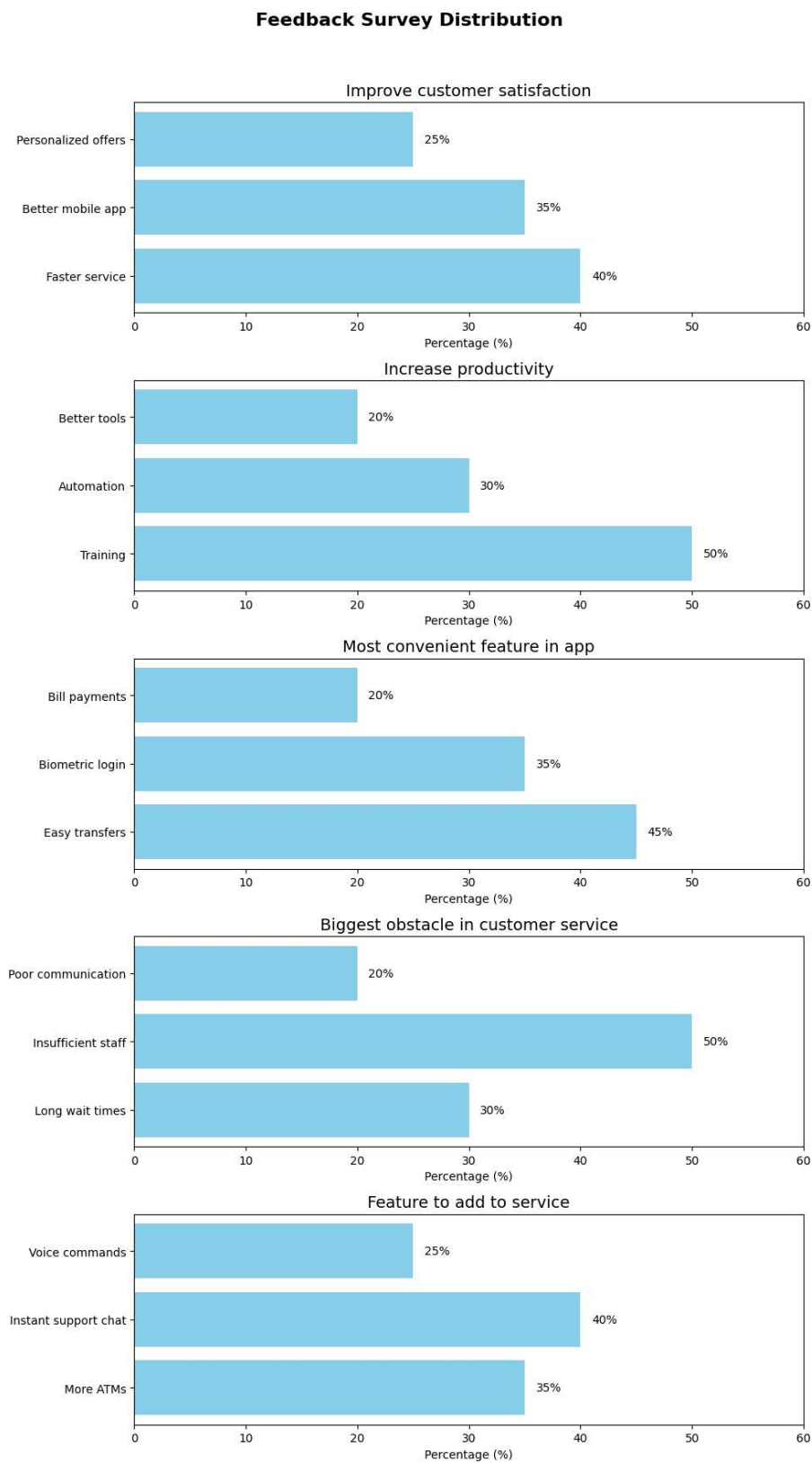


Figure 3.13: Comparative Analysis of Responses Across Multiple Categories.

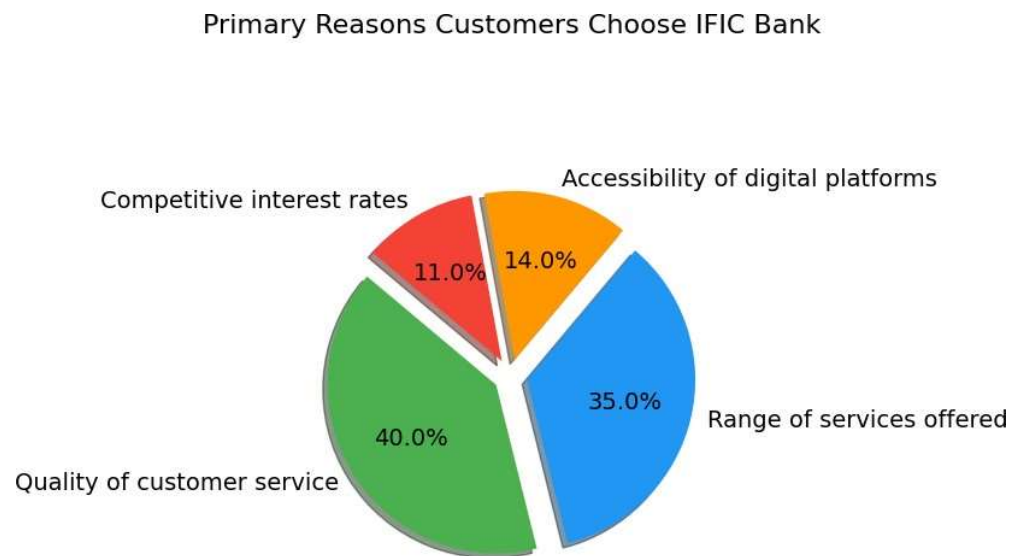


Figure 3.14: Distribution of Primary Reasons Customers Prefer IFIC Bank over Competitors.

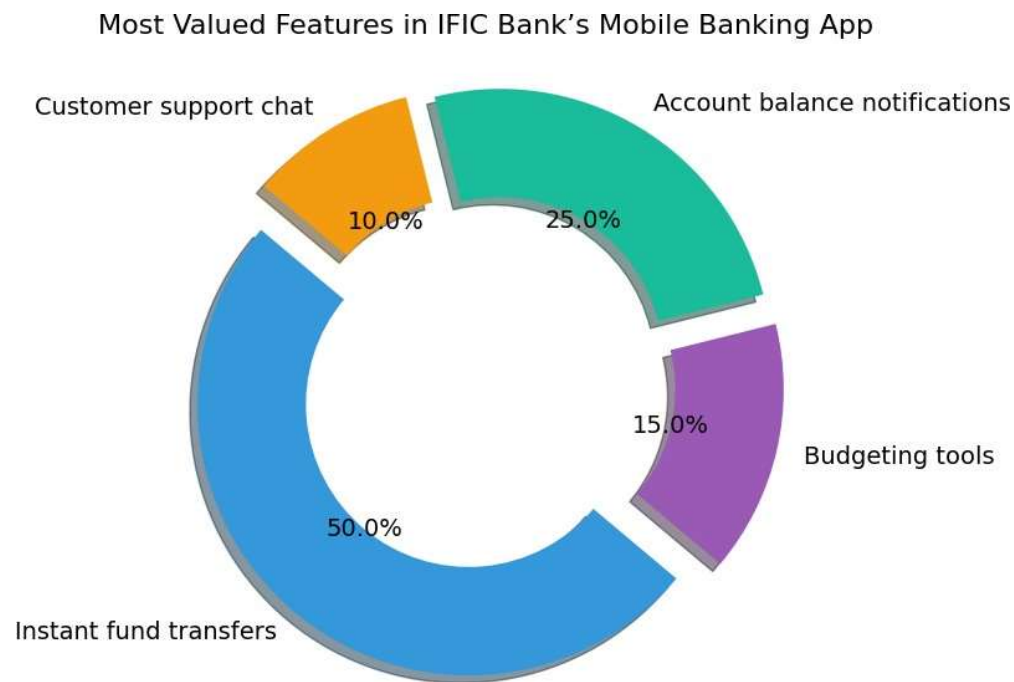


Figure 3.15: Distribution of Most Valued Features in IFIC Bank’s Mobile Banking App.

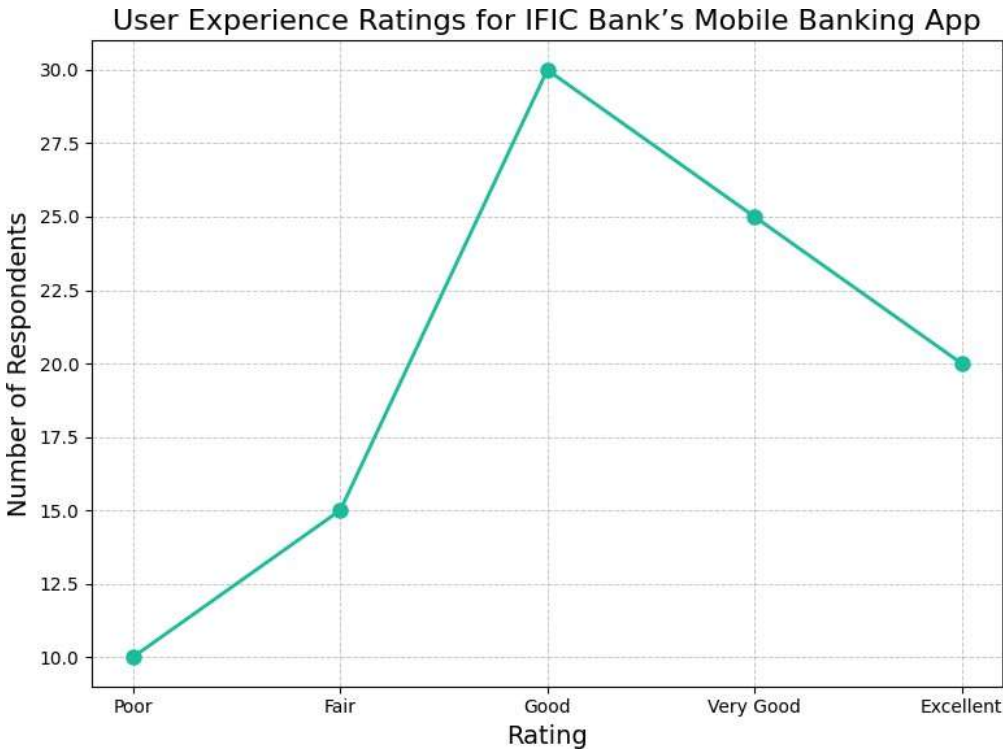


Figure 3.16: Illustration of User Experience Ratings for IFIC Bank's Mobile Banking App.

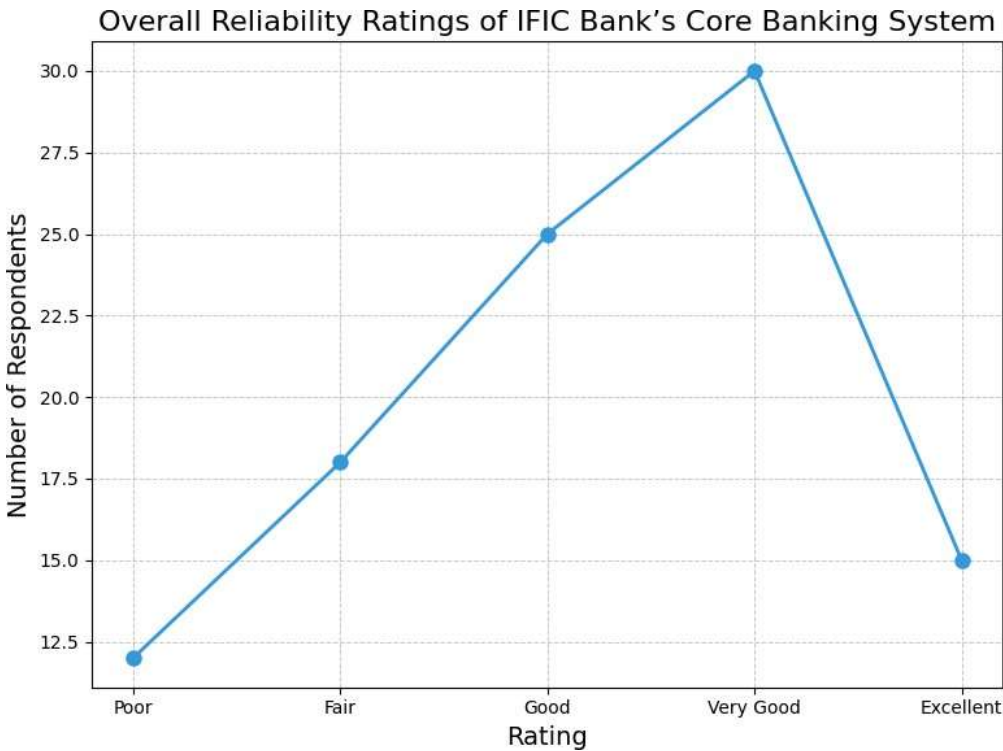


Figure 3.17: Illustration of Overall Reliability Ratings of IFIC Bank's Core Banking System.

3.3.2 DFDs of Existing System

Data Flow Diagrams (DFDs) are crucial tools in system analysis and design, used to visually represent the flow of data within a system. They provide a structured approach to understanding how data moves through various processes, how it is stored, and how it interacts with external entities. We have designed the workflow of current system that is aligned with the problems we identified.

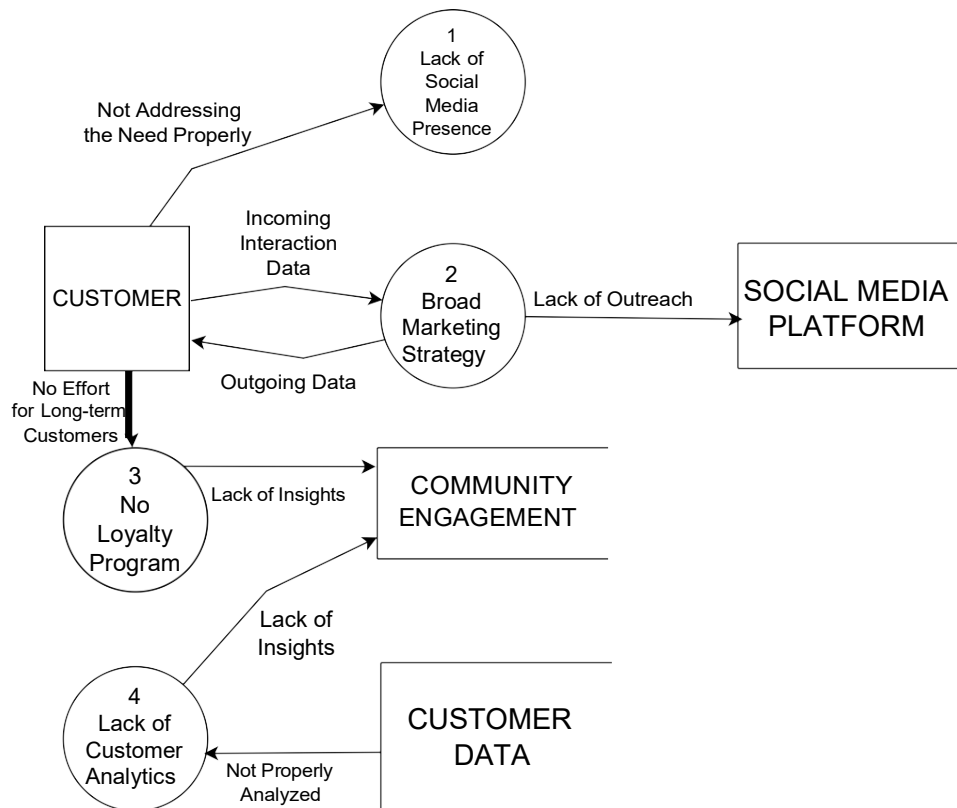


Figure 3.18: DFD showing Problem 1: Inadequate Marketing and Customer Outreach.

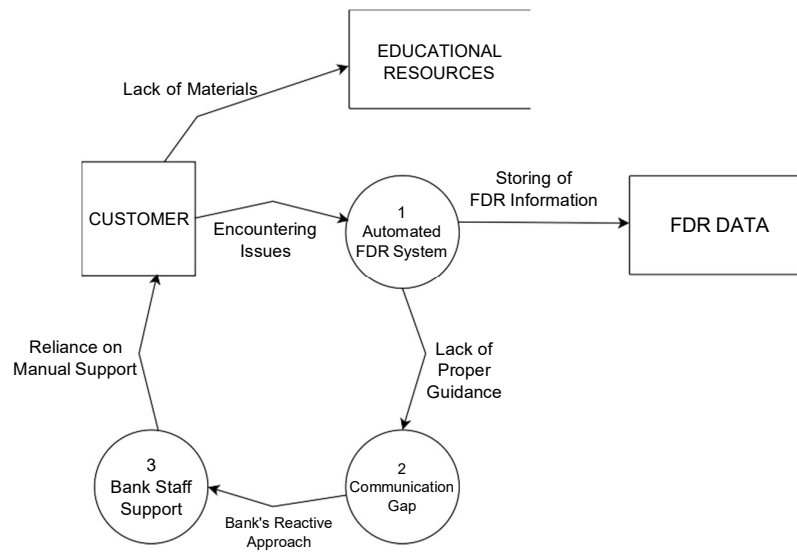


Figure 3.19: DFD showing Problem 2: Lack of Credibility with Automated Fixed Deposit Receipt (FDR) System.

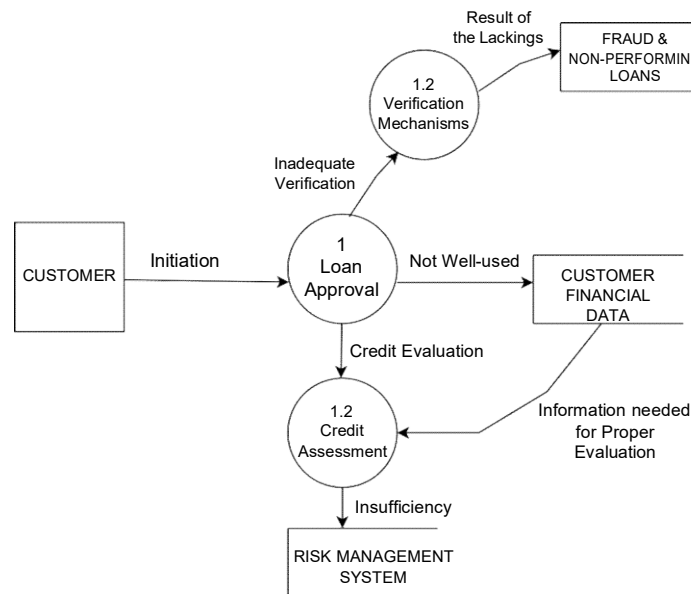


Figure 3.20: DFD showing Problem 3: Inadequate Customer Background Checks in Loan Approval.

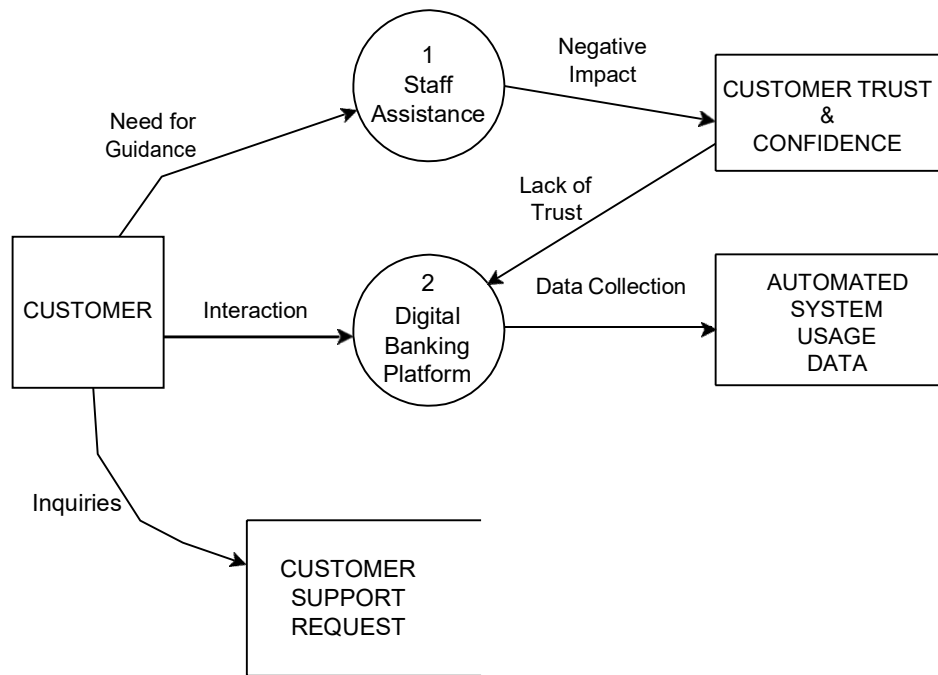


Figure 3.21: DFD showing Problem 4: Insufficient Manpower and Staffing Issues.

3.4 Detailed Feasibility Analysis

This section provides a comprehensive examination of the root causes of the key issues identified at IFIC Bank PLC, Rajshahi Branch. We explore the problems in detail, propose three candidate systems to address these challenges, and evaluate each system based on performance, cost, and effectiveness. The analysis concludes by selecting the most viable system for implementation.

3.4.1 Root Cause Analysis of Identified Problems

The problems at IFIC Bank PLC, Rajshahi Branch can be grouped into four primary categories: inadequate marketing and customer outreach, credibility issues with the automated Fixed Deposit Receipt (FDR) system, inadequate customer background checks in loan approvals, and insufficient staffing for customer support in digital services. Each of these issues is critical to the bank's overall performance, customer satisfaction, and growth potential.

1. Inadequate Marketing and Customer Outreach

Root Cause: The marketing strategies employed by the Rajshahi Branch are outdated, non-targeted, and lacking in digital engagement. This deficiency has resulted in poor visibility, low customer acquisition rates, and ineffective brand presence. Traditional marketing efforts have failed to leverage the power of data analytics, social media, and targeted campaigns, causing the branch to miss opportunities to engage key customer segments.

Performance Impact: Without an effective marketing strategy, the branch has struggled to attract new customers, particularly in underserved areas where competitors have seized market share. Younger, tech-savvy individuals are particularly underrepresented in the branch's customer base due to its poor digital presence.

Behavioral Impact: Customers, especially younger generations, prefer digital interactions over traditional banking methods. By failing to engage these customers through modern platforms, the branch is not only limiting its growth but also risking losing customers to more digitally advanced competitors.

2. Credibility Issues with the Automated FDR System

Root Cause: The automated FDR system at the branch suffers from credibility issues due to a lack of transparency in communication, discrepancies between expected and actual terms, and insufficient guidance provided to customers during the FDR creation process. Many customers have reported mistrust in the system, which hampers adoption rates.

Performance Impact: Credibility issues have resulted in a low adoption rate for the automated FDR system, reducing the number of deposits made through this channel. This also impacts customer retention, as trust in the bank's automated systems is crucial for maintaining long-term relationships.

Technical Impact: The FDR system's technical limitations prevent it from providing real-time updates or confirmations, creating a sense of uncertainty among customers. Additionally, the lack of a streamlined interface exacerbates the problem for customers who are not familiar with digital banking platforms.

3. Inadequate Customer Background Checks in Loan Approvals

Root Cause: The Rajshahi Branch's loan approval process is hindered by insufficient customer background checks, which have led to increased loan defaults. Manual and outdated methods for verifying customer creditworthiness and financial history have resulted in poor risk management and financial losses for the bank.

Performance Impact: The inadequacy of the current loan approval process has increased the branch's exposure to high-risk customers. The lack of automated checks for creditworthiness, debt-to-income ratios, and financial histories has resulted in a higher rate of loan defaults, undermining the branch's profitability.

Economic Impact: Financial losses due to defaults significantly affect the branch's bottom line. The lack of thorough background checks and automated risk assessment processes makes the bank vulnerable to fraudulent or high-risk loan applications.

4. Insufficient Staffing and Customer Support

Root Cause: Despite the bank's increasing focus on digital services, the Rajshahi Branch is understaffed, particularly in customer support roles. This limits the bank's ability to guide customers through the transition to digital services, leading to a lack of digital adoption and customer dissatisfaction.

Performance Impact: Insufficient customer support affects the adoption of digital banking solutions. Many customers, particularly older individuals or those unfamiliar with technology, struggle to navigate online platforms without adequate assistance. This gap in service delivery results in under-utilization of the branch's digital infrastructure.

Operational Impact: The existing staff are often overwhelmed with assisting customers who are not comfortable with the digital systems, slowing down overall service delivery. This issue also reduces the efficiency gains expected from digital transformation initiatives.

3.4.2 Candidate Systems for Resolving Identified Issues

In light of the root cause analysis, we propose three candidate systems to address the issues at IFIC Bank PLC, Rajshahi Branch. Each system is designed to tackle one or more of the problems identified, and will be evaluated based on technical, economic, and behavioural feasibility.

Candidate System 1: Integrated Customer Relationship Management (CRM) and Marketing Platform

This system is specifically designed to address the complexities associated with marketing and customer outreach in today's dynamic business environment. By combining advanced data analytics with a comprehensive Customer Relationship Management (CRM) platform, the system enables organizations to implement highly targeted marketing campaigns, streamline customer interactions, and enhance customer support.

Furthermore, the integration of data analytics tools enables the continuous monitoring of customer interactions, providing valuable insights into customer behavior and preferences. This information is crucial for designing effective marketing strategies that resonate with each customer group, thereby increasing engagement and satisfaction. The CRM platform not only supports personalized marketing initiatives but also facilitates efficient customer support by centralizing customer information and interaction history, allowing staff to respond promptly and effectively to customer inquiries.

Characteristics:

- **Focus:** The primary focus of this system is to leverage data-driven marketing strategies, effective customer segmentation, and streamlined support processes to enhance overall customer engagement and satisfaction. The emphasis is on understanding customer needs and delivering tailored solutions that foster long-term relationships.
- **Technology Requirements:** To operate effectively, this system requires seamless integration with various technology platforms, including customer databases, analytics tools, and marketing platforms such as email and SMS services. The CRM software must be capable of handling large volumes of customer data while providing robust analytics capabilities to derive actionable insights.
- **Benefits:** The implementation of this integrated CRM and marketing platform offers numerous benefits, including improved customer acquisition rates. By utilizing targeted marketing strategies, organizations can expect higher engagement levels and increased brand visibility. The centralized approach to customer support leads to a more efficient handling of customer inquiries, ultimately enhancing the overall customer experience.

Candidate System 2: Advanced Risk Management and Loan Processing System

The Advanced Risk Management and Loan Processing System is designed to revolutionize the loan approval process by introducing a fully automated mechanism for assessing creditworthiness and managing risk. This system integrates comprehensive customer credit checks, income verification processes, and risk assessment tools powered by advanced data analytics. By providing real-time feedback, it aims to ensure a streamlined, secure, and efficient loan approval experience, directly addressing challenges related to inadequate background checks and risk management.

In addition to expediting the approval process, the system ensures that customers receive real-time notifications regarding the status of their loan applications. This immediate feedback allows applicants to remain informed throughout the process, thereby enhancing their overall experience and satisfaction with the banking institution.

Characteristics:

- **Focus:** The primary focus of the system is on automating risk assessment and credit checks, which leads to streamlined loan approvals and improved decision-making processes. By implementing these technologies, the system enhances the accuracy and reliability of loan evaluations, ensuring that only qualified applicants receive funding.
- **Technology Requirements:** To operate effectively, this system requires seamless integration with credit bureaus for obtaining real-time credit scores and financial data. It also relies on advanced data analytics tools, automated credit scoring algorithms, and robust risk management systems to facilitate comprehensive assessments.
- **Benefits:** The implementation of the Advanced Risk Management and Loan Processing System is expected to yield several significant benefits. These include a marked reduction in loan defaults due to improved risk assessment practices, enhanced decision-making capabilities, and significantly faster loan processing times. Ultimately, these enhancements contribute to increased customer satisfaction, as applicants can enjoy a smoother and more transparent loan approval journey.

Candidate System 3: Comprehensive Digital Banking System with Enhanced Customer Support

This comprehensive digital banking system is designed to upgrade existing banking infrastructure by making it more transparent, user-friendly, and reliable. The system integrates key features such as an automated Fixed Deposit Receipt (FDR) process, real-time notifications, and an AI-driven support assistant. The primary goal of these enhancements is to improve customer interactions and address common issues related to credibility and staffing in banking services.

A critical component of this system is the automated FDR process, which simplifies the application and management of fixed deposit accounts. Customers will benefit from real-time updates regarding FDR terms, interest rates, and any changes that may occur, thereby ensuring they are well-informed and able to make timely decisions. This transparency in banking processes helps build trust and confidence among customers, as they have access to accurate information regarding their investments.

In addition to the automated FDR features, the system incorporates an AI-driven support assistant that provides immediate assistance for customers navigating digital banking services. By offering real-time support for various inquiries, customers can resolve issues quickly without needing to wait for human staff intervention. This capability not only enhances the overall customer experience but also alleviates the pressure on human resources, allowing staff to focus on more complex customer needs.

Characteristics:

- **Focus:** The system emphasizes enhancing digital banking services by providing real-time updates and improving customer support. The integration of transparent processes ensures that customers are consistently informed about their banking activities, fostering a sense of reliability and confidence in the services offered.
- **Technology Requirements:** To function optimally, this system necessitates integration with legacy banking systems, which may include traditional databases and transaction processing systems. Additionally, it requires robust real-time data processing capabilities, AI-driven support tools, and automated notification systems for communicating updates

through various channels such as email and SMS.

- **Benefits:** The implementation of this comprehensive digital banking system is poised to yield numerous benefits. Customers can expect increased trust in the bank's services due to enhanced transparency and real-time information. Furthermore, the higher adoption rate of digital services will likely lead to reduced strain on human staff, as the AI-driven assistant can handle a significant volume of routine inquiries. Ultimately, these improvements will contribute to elevated overall customer satisfaction, establishing a more responsive and efficient banking environment.

3.4.3 Evaluation of Candidate Systems

Each candidate system will be evaluated based on its performance, cost-effectiveness, technical feasibility, and potential impact on the branch's operations.

Performance and Cost Effectiveness

Candidate System 1: Integrated Customer Relationship Management (CRM) and Marketing Platform

Performance: This system will provide a centralized platform for managing customer interactions, marketing campaigns, and customer feedback. It will include features like customer segmentation, targeted digital marketing, automated follow-ups, and a customer support portal. By enhancing both marketing and customer support, this system addresses inadequate marketing strategies and customer outreach issues. It can also help the branch monitor customer satisfaction and tailor services to meet customer needs.

Cost Effectiveness: Implementing a CRM with integrated marketing tools has an upfront cost, but it allows for streamlined customer interactions, reducing the need for excessive manpower. The CRM can automate customer outreach, reducing manual workload and improving marketing efficiency. Improved customer engagement and satisfaction can drive customer acquisition, retention, and cross-selling, increasing the bank's revenue in the long term.

Potential Risks: The system requires staff training for effective use, and initial implementation can be complex. Resistance to adopting new digital tools by both staff and customers may slow down its effectiveness, requiring a strategic change management plan.

Candidate System 2: Advanced Risk Management and Loan Processing System

Performance: This system will employ advanced risk assessment algorithms and data analytics for loan processing, ensuring thorough background checks and reducing loan default risks. It will integrate with the CRM to gather customer insights and use automated credit scoring to streamline loan approvals. The system will also feature real-time monitoring of loan performance and proactive notifications to customers about repayment schedules. By automating risk assessments and monitoring, the bank can manage loans more efficiently and address staffing issues, reducing reliance on manual background checks.

Cost Effectiveness: Although implementing an advanced risk management system involves a high initial investment, the system will significantly decrease loan defaults, saving the bank from potential losses. Automated credit assessments will quicken the loan approval process, improving customer satisfaction and potentially increasing loan volumes. This will ultimately result in a higher return on investment.

Potential Risks: The high cost of setup and maintenance, as well as data security, are major concerns. The bank must implement stringent data protection measures to safeguard customer information. Additionally, a robust strategy for staff training and customer education will be crucial for smooth adoption.

Candidate System 3: Comprehensive Digital Banking System with Enhanced Customer Support

Performance: This system will upgrade the current digital banking infrastructure to provide a more user-friendly, transparent, and reliable experience. It will integrate enhanced automated FDR processes, real-time notifications, and a more straightforward loan application procedure. To address the customer support gap, the system will feature a built-in, AI-driven support assistant for immediate help and a dedicated hotline for personalized assistance. This approach tackles issues of credibility in the FDR system, inadequate customer support due to insufficient manpower, and customer hesitancy toward digital banking services.

Cost Effectiveness: While the initial setup and enhancement of the digital banking system may involve moderate to high costs, these expenses will be offset by long-term gains from increased customer adoption of digital services. Automated support tools will reduce the strain

on staff, allowing them to focus on more complex customer issues. Improved customer trust and satisfaction will enhance the bank’s reputation, encouraging more deposits and loan applications.

Potential Risks: Integration with legacy systems and ensuring seamless customer experiences may be challenging. Some customers, especially those unfamiliar with technology, may still require additional support, necessitating a gradual adoption strategy and dedicated training resources.

The “Candidate Qualitative Evaluation Matrix” provides a detailed comparison of three systems based on key performance criteria like technical feasibility, customer satisfaction, operational impact, and long-term ROI, using qualitative terms such as “High” and “Very High.” This table simplifies understanding which system excels in specific areas without delving into numerical values.

Table 3.1: Candidate Qualitative Evaluation Matrix

Evaluation Criteria	Candidate System 1	Candidate System 2	Candidate System 3
Technical Feasibility	High	High	Very High
Cost Effectiveness	Very High	High	Very High
Customer Satisfaction	High	Very High	High
Implementation Time	High	Medium	High
Operational Impact	High	High	Very High
Long-term ROI	Very High	High	Very High

The “Candidate System Performance/Cost Evaluation Matrix” offers a more granular breakdown of both performance and cost criteria, showing subtopics like cost-effectiveness and implementation time. It incorporates real implementation times in months and uses percentages to highlight where each system performs best, offering a clearer comparison of overall system suitability based on both qualitative and time-based metrics.

3.4.4 Weighting

System Performance and Cost Data

Each candidate system is evaluated based on a weighted scoring model that considers key criteria such as technical feasibility, implementation cost, customer satisfaction, and long-term benefits. Weighting these criteria allows for a more objective assessment, with greater emphasis placed

Table 3.2: Candidate System Performance/Cost Evaluation Matrix (with Breakdown and Percentages)

Evaluation Criteria	Candidate System 1	Candidate System 2	Candidate System 3
Performance Breakdown			
Technical Feasibility	High (88%)	High (89%)	Very High (100%)
Customer Satisfaction	High (89%)	Very High (100%)	High (89%)
Operational Impact	High (89%)	High (78%)	Very High (100%)
Long-term ROI	Very High (100%)	High (89%)	Very High (100%)
Cost Breakdown			
Cost Effectiveness	Very High (100%)	High (88%)	Very High (100%)
Implementation Time	6 months (100%)	7 months (85%)	6 months (100%)
Overall Performance/Cost Evaluation			
Overall Performance	High (91%)	High (89%)	Very High (100%)
Overall Cost	Medium (95%)	High (87%)	Medium (95%)

on factors like customer satisfaction and long-term benefits, as they directly impact the bank’s growth and stability. For example, higher weights might be assigned to customer satisfaction and risk management due to their critical roles in building trust and ensuring sustainable operations. This comprehensive evaluation ensures that the chosen system aligns with the bank’s strategic goals and maximizes return on investment.

3.4.5 Selecting the Best Candidate System

Based on the weighted evaluation, **Candidate System 3 (Comprehensive Digital Banking System with Enhanced Customer Support)** emerges as the best option due to its ability to provide the most substantial long-term benefits. This system not only reduces loan defaults through improved risk management but also enhances overall operational efficiency. By integrating advanced digital banking features with robust customer support, it addresses multiple core problems simultaneously, including customer satisfaction, transparency, and effective use of staff resources. This comprehensive approach makes it the most strategic and impactful solution for implementation.

Table 3.3: Weighted Candidate Evaluation Matrix

Evaluation Criteria	Weight Factor	Candidate System 1		Candidate System 2		Candidate System 3	
		Rating	Score	Rating	Score	Rating	Score
Performance							
Technical Feasibility	8	7	56	8	64	9	72
Cost Effectiveness	7	8	56	7	49	9	63
Customer Satisfaction	9	8	72	9	81	8	72
Costs							
Implementation Time	6	7	42	6	36	7	42
Operational Impact	8	8	64	7	56	9	72
Long-term ROI	10	9	90	8	80	9	90

3.5 Cost-Benefit Analysis

Cost-benefit analysis is a systematic method that organizations use to estimate the potential costs and benefits of various decisions, systems, or projects. This process helps to identify the most efficient and effective alternatives. By providing a comprehensive assessment of the outcomes associated with a particular action, cost-benefit analysis becomes an essential tool for formulating business strategies, assessing new hires, and making informed decisions on resource allocation or purchases. In this report, we developed a cost-benefit analysis model by clearly identifying the benefits of a given decision, calculating the associated costs, and determining the net result by subtracting those costs from the benefits.

3.5.1 Costs and Benefits Identification

Costs and benefits identification is important to construct the most cost-effective model from the data. Some costs were easily identified or quantified while others were hard to determine.

In any system implementation, it is crucial to thoroughly identify and classify the associated costs to ensure a comprehensive cost-benefit analysis. For the IFIC Bank system, costs can be divided into two primary categories: tangible costs and intangible costs.

Tangible Costs

Tangible costs are easily quantifiable and directly associated with the project. These are the measurable financial outlays that the bank will incur during the system implementation. The key tangible costs include:

- **Initial Setup Costs:** These include expenses for hardware and software purchases, installation fees, and initial setup activities required for system deployment. For this system, the setup costs are estimated at BDT 1,000,000.
- **Monthly Operational Costs:** These are the recurring costs involved in maintaining the system after its implementation. They cover system maintenance, employee salaries, utilities, and other running costs. For this analysis, monthly operational costs are projected at BDT 100,000.
- **Security Upgrades:** An additional tangible cost is the upgrade of security measures, which is necessary to ensure the system's compliance with banking regulations and pro-

tection against cyber security threats. This is estimated at BDT 50,000.

- **Annual License Fee:** The bank must pay an annual fee for software licenses required to run the system, which amounts to BDT 200,000.
- **Consultancy Fees:** Engaging external consultants for system integration and technical support is another tangible cost. The consultancy fee is estimated at BDT 150,000.

Intangible Costs

Intangible costs are those that are harder to quantify but still have a significant impact on the project's overall success. These costs are often related to human factors, productivity, and customer experience. The key intangible costs include:

- **Disruption During System Transition:** The transition from the old system to the new one may temporarily disrupt banking operations. This could lead to delays, reduced service efficiency, and customer dissatisfaction during the implementation phase.
- **Employee Training and Adaptation:** Staff members will need time and resources to learn the new system, which could result in decreased productivity during the initial phases. The cost associated with training and adaptation is challenging to measure directly but can impact overall efficiency.
- **Customer Dissatisfaction:** During the transition period, customers may experience delays or interruptions in services, leading to short-term dissatisfaction. While this is difficult to quantify financially, it could affect customer loyalty and reputation.
- **Productivity Loss:** During the initial phase of system adoption, there may be a temporary decline in employee productivity as they adapt to the new processes, which could affect overall performance.
- **Long-Term Customer Service Impact:** Although the new system is designed to improve long-term customer service, there may be challenges in the short term as employees become proficient in using the new tools and technologies.

By identifying both the tangible and intangible costs, the organization can better assess the full financial and operational impact of the system implementation and make informed decisions based on this comprehensive understanding.

Tangible and Intangible Cost Breakdown Costs for the system implementation can be divided into two categories: tangible and intangible.

Table 3.4: Tangible and Intangible Cost Breakdown

Cost Type	Tangible Costs (USD)	Intangible Costs
Initial Setup Cost	BDT 1,000,000 (Hardware, software, setup)	Potential disruption during system switch
Monthly Operational Costs	BDT 100,000 (Maintenance, salaries, utilities)	Employee training and adaptation
Other Costs	Security upgrades, BDT 50,000	Customer dissatisfaction during transition
Annual License Fee	BDT 200,000	Impact on employee productivity
Consultancy Fee	BDT 150,000 (for integration consultants)	Long-term customer service impact

3.5.2 Benefits of System Implementation

1. **Enhanced Efficiency and Automation:** A robust banking system streamlines processes and automates routine tasks, significantly boosting operational efficiency. By reducing manual efforts, employees can focus on complex, value-added activities, improving overall productivity and minimizing errors or delays in transaction processing.
2. **Improved Customer Experience:** A well-designed system enhances customer satisfaction by providing user-friendly interfaces, allowing for easy account access, quick transactions, and real-time updates. Faster service delivery, coupled with personalized services based on customer preferences, strengthens loyalty and improves the overall customer experience.
3. **Risk Management and Compliance:** Advanced banking systems incorporate sophisticated algorithms for risk management, detecting and preventing fraudulent activities. Enhanced security ensures the protection of customer accounts and transactions, while built-in compliance features help banks adhere to regulatory standards, reducing the risk of penalties and reputational damage.
4. **Data Analytics and Business Intelligence:** Modern systems include powerful data analytics tools that process vast datasets, offering valuable insights for data-driven decision-making. By identifying trends and customer behavior, banks can optimize operations, target marketing strategies, and explore new business opportunities.
5. **Competitive Advantage:** Implementing cutting-edge banking technologies, such as

mobile banking and personalized services, sets a bank apart from its competitors. Staying ahead of technological advancements helps attract new customers, retain existing ones, and create a seamless digital experience, fostering a competitive edge in the market.

6. **Scalability and Future Growth:** A flexible banking system is scalable, capable of handling increasing transaction volumes, customer growth, and expanding product lines. This scalability minimizes the need for frequent upgrades, reduces costs, and ensures smooth operations as the bank expands.

7. **Cost Reduction and Revenue Generation:** Automation and efficiency improvements reduce operational costs by eliminating redundant tasks and optimizing resource allocation. With accurate data analytics, banks can identify cross-selling opportunities, launch targeted marketing campaigns, and introduce fee-based services, generating additional revenue streams.

A modern banking system offers enhanced efficiency, improved customer experiences, stronger risk management and compliance, data-driven decision-making, a competitive advantage, scalability for growth, and opportunities for cost reduction and revenue generation. These benefits contribute to the bank's long-term success and competitiveness in the rapidly evolving financial sector.

3.5.3 Break Even Analysis

3.5.3.1 Break-Even Point

To calculate the break-even point (BEP), we use the following formula:

$$\text{Break-Even Point (BEP) in units} = \frac{\text{Fixed Costs}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$

Fixed and Variable Costs

- **Fixed Costs:** These are costs that remain constant regardless of how much is produced or sold.
 - Initial Setup Cost: BDT 1,000,000
 - Other Costs (Security upgrades): BDT 50,000

- Annual License Fee: BDT 200,000
- Consultancy Fee: BDT 150,000

Total Fixed Costs = BDT 1, 000, 000+BDT 50, 000+BDT 200, 000+BDT 150, 000 = BDT 1, 400, 000

- **Variable Costs:** These are costs that change with the level of production.
 - Monthly Operational Costs: BDT 100,000 (treated as a variable cost over time depending on the production structure).

Selling Price and Variable Cost per Unit

- Selling Price per Unit = BDT 500
- Variable Cost per Unit = BDT 100

Break-Even Point Calculation

$$\text{BEP (in units)} = \frac{1,400,000}{500 - 100} = \frac{1,400,000}{400} = 3,500 \text{ units}$$

The break-even point is 3,500 units. This means the bank must sell 3,500 units to cover its total fixed and variable costs.

3.5.3.2 Break-Even Revenue

To calculate the total revenue needed to break even, we multiply the break-even point by the selling price per unit:

$$\text{Break-Even Revenue} = \text{BEP (units)} \times \text{Selling Price per Unit}$$

$$\text{Break-Even Revenue} = 3,500 \times 500 = \text{BDT } 1,750,000$$

The break-even revenue is BDT 1,750,000.

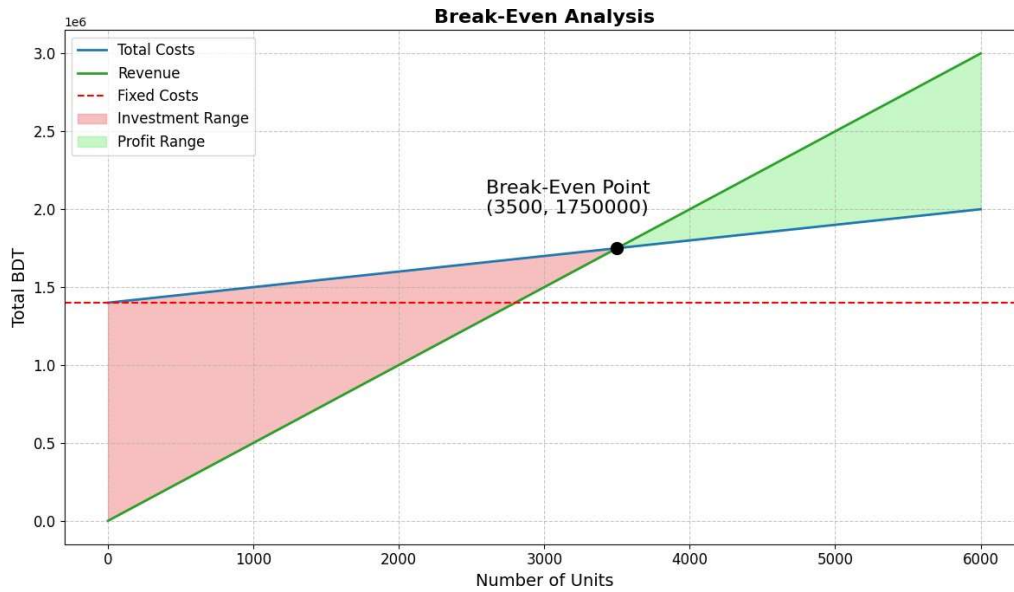


Figure 3.22: Break-Even Chart of the Candidate System.

3.6 Conclusion

We discussed various information gathering tools and used these tools to collect the necessary information about IFIC Bank PLC. In this process, we covered all the services and systems of the bank. We also mentioned our official conversation with the Branch Manager of the Rajshahi branch of IFIC Bank PLC and gathered all the information we needed. Afterward, we provided a summary of the data collected through online sources and interviews. Finally, we developed some Data Flow Diagrams (DFD) for IFIC Bank's existing system that highlights our found problems, organized in a structured way with various steps to make it easier to understand and analyze.

Chapter 4

Design

4.1 Introduction

In systems development, systems design is a critical process that transforms the requirements identified during systems analysis into a detailed blueprint for building a functional system. This phase involves defining the system's architecture, components, and data flow to meet user requirements. A key element of systems design is the creation of input methods and output reports that are both user-friendly and efficient. Properly designed inputs, often in the form of data collection forms, and well-structured outputs ensure the system functions seamlessly and aligns with user expectations.

Forms play a vital role in organizations, serving as tools to capture the precise data needed for system operations. Without accurate data, a system cannot function effectively. Forms help automate data collection, saving time and improving the accuracy of reports by capturing the necessary information required by organizational members. Each form represents specific data needs and must be designed or updated accordingly to ensure smooth system integration.

Alongside forms, databases are another crucial component in systems design. Databases organize and process data, establishing relationships between various pieces of information, allowing multiple users and applications to access data easily and quickly. Together, the design of forms and databases ensures that the system not only meets organizational goals but also operates efficiently, providing reliable data and facilitating informed decision-making.

4.2 Data Flow Diagram of the Proposed System

IFIC Bank PLC, Rajshahi Branch, is committed to enhancing its digital banking system to address key operational challenges, including credibility gaps in the automated Fixed Deposit Receipt (FDR) system, insufficient customer support, and inefficiencies in loan approval. The proposed solution, titled Comprehensive Digital Banking System with Enhanced Customer Support, focuses on resolving these issues by integrating AI-driven features, real-time notifications, and improved customer interactions. This section outlines the changes from the existing system, emphasizing automation, transparency, and enhanced customer engagement.

The current digital banking system faces several limitations. The proposed system will address these through data-driven customer insights, enhanced automation, and a focus on long-term customer relationships. The primary changes include improved FDR automation, AI-driven customer support, real-time communication, and an enhanced loan approval process.

4.2.1 Key Improvements in Customer Engagement

One of the major issues identified in the existing system is the lack of effective customer outreach. The new system introduces advanced customer analytics to personalize services and outreach strategies, enhancing customer retention through AI-driven assistance, a loyalty program, and a dedicated support hotline. By employing these tools, the bank can offer personalized support and engage more effectively through community platforms and social media.

AI-Driven Support and Automation

The new AI-driven support system automates responses to customer inquiries, reducing delays in issue resolution. This eliminates manual interventions and improves overall efficiency. Additionally, the loyalty program aims to build stronger relationships with long-term customers, providing incentives for continued engagement.

Community Engagement

Insights gained from the customer analytics system inform engagement strategies, ensuring that outreach efforts are aligned with customer preferences and behaviors. This not only improves the marketing effectiveness but also fosters better relationships through tailored communication.

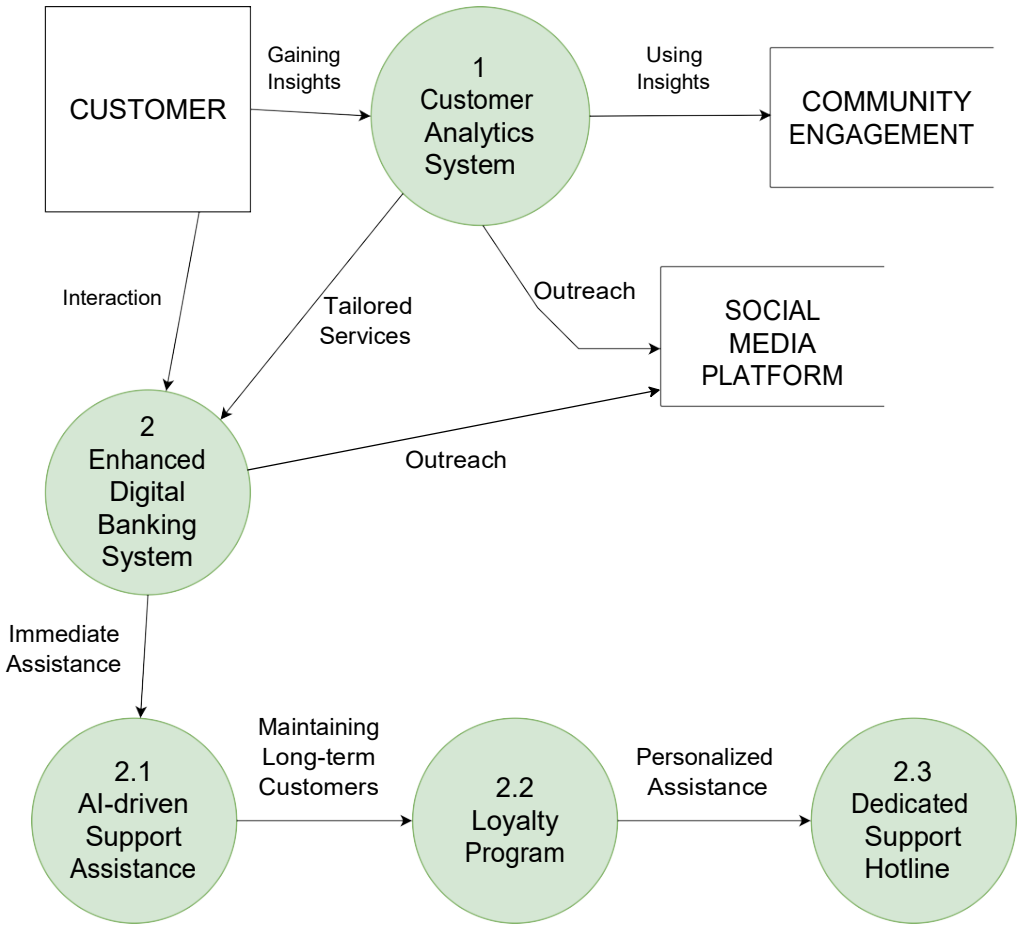


Figure 4.1: Proposed DFD for Comprehensive Customer Outreach. (Changes are highlighted in green.)

4.2.2 Addressing Credibility in FDR Automation

The existing automated Fixed Deposit Receipt (FDR) system suffers from credibility issues due to a lack of transparency and delayed responses. The proposed solution focuses on improving customer trust by introducing real-time communication, proactive error resolution, and comprehensive educational resources for users. This ensures that customers are well-informed and have access to immediate support when needed.

Proactive Issue Resolution

In contrast to the reactive nature of the current system, the proposed FDR process is designed to identify and resolve potential problems before they affect the customer. This shift improves system reliability and customer satisfaction, while minimizing disruptions.

Educational Resources

To empower customers, the system includes detailed resources, helping users navigate the platform with ease and reducing their dependency on bank staff. This fosters a more self-sufficient customer base, reducing the need for support interventions.

AI-Driven Support Assistance

The introduction of AI-driven support provides immediate and automated customer assistance. This system significantly reduces the need for manual intervention by bank staff, streamlining the resolution of customer queries related to Fixed Deposit Receipts (FDR). By leveraging AI, the support system can handle common issues more efficiently, enhancing the overall customer experience and ensuring faster response times.

Real-Time Communication

Real-time communication ensures that customers receive timely responses to their inquiries, eliminating delays and confusion. This feature is essential for providing instant guidance, particularly during the FDR process, where transparency and immediate support are crucial. By offering real-time interaction, the system enhances customer trust and satisfaction, helping to address credibility concerns associated with the previous system.

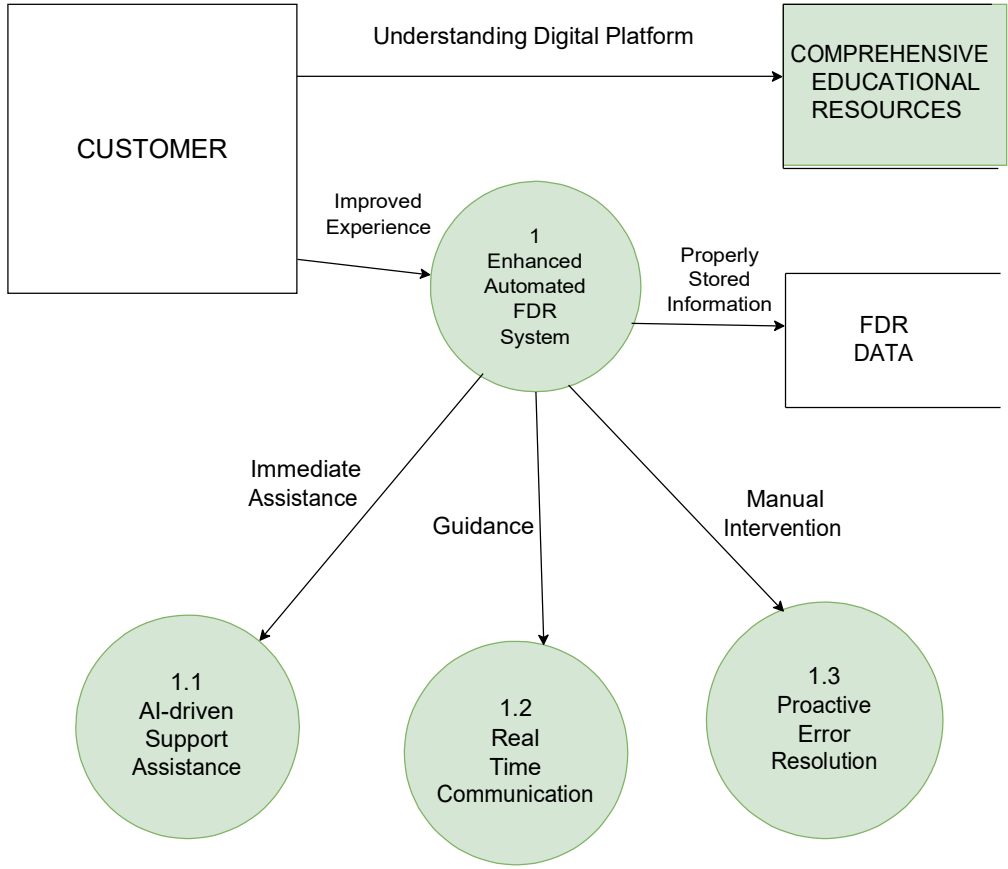


Figure 4.2: Proposed DFD of the Enhanced FDR System.(Changes are highlighted in green.)

4.2.3 Enhancements in Loan Approval Process

One of the major concerns identified in the current system is the insufficient background checks for loan applicants. The proposed system addresses this by implementing automated verification mechanisms that assess a customer's financial data, creditworthiness, and associated risks in real time.

Automated Verification Mechanism

The new system introduces an automated verification process that ensures accurate validation of customer financial data. By automating this step, the system reduces the likelihood of errors or fraudulent data slipping through the approval process. This enhancement drastically improves the speed and reliability of loan applications, minimizing manual intervention while ensuring compliance with verification standards. Moreover, this automated verification integrates seamlessly with external data sources, such as credit bureaus and financial regulators, ensuring real-time access to accurate customer data, which reduces the time required for approvals and mitigates risks associated with incomplete or outdated information.

Advanced Risk Management Tools

To complement the verification process, the system integrates advanced risk management tools. These tools assess a customer's creditworthiness and financial history, ensuring that only low-risk candidates receive loan approvals. By analyzing a comprehensive set of data points, including spending behavior, repayment history, and financial obligations, the system minimizes the chances of granting loans to high-risk individuals. Additionally, the system employs predictive analytics to anticipate potential financial risks, offering early warnings for high-risk loan applicants. This risk management strategy not only reduces the likelihood of non-performing loans but also strengthens the bank's overall loan portfolio, contributing to long-term financial stability and sustainability.

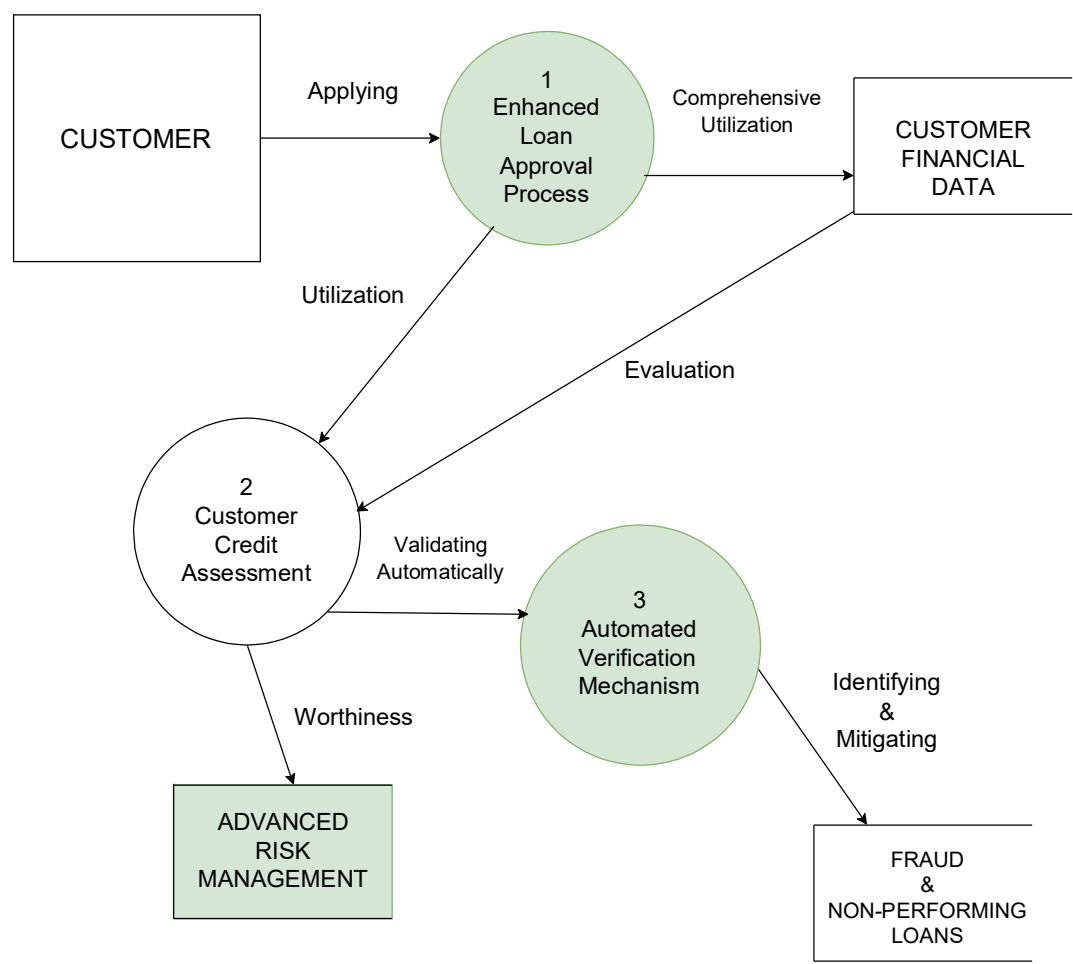


Figure 4.3: Proposed DFD of the Loan Approval System.(Changes are highlighted in green.)

4.2.4 Automation to Address Staffing Shortages

Staffing shortages and dependency on manual interventions have hindered the efficiency of customer support in the current system. The proposed enhancements include automation in customer service, which significantly reduces the workload on bank staff and ensures faster resolutions for customer queries.

Enhanced Customer Support System

The enhanced customer support system is designed to elevate the overall customer experience by providing a structured approach to support requests. This system streamlines communication channels, ensuring that customer inquiries are routed efficiently to the appropriate resources. By incorporating features such as a dedicated support hotline and an online help center, the system empowers customers to access assistance whenever needed. Additionally, the support system is designed to analyze common queries and trends, enabling proactive outreach and better service personalization.

Automated Assistance Mechanism

The automated assistance mechanism plays a crucial role in handling routine inquiries and support tasks. By leveraging AI technology, this mechanism can provide instant responses to frequently asked questions, enabling faster resolution times and reducing the need for human intervention. This automation ensures that customers receive immediate help for basic issues, allowing bank staff to concentrate on more complex queries that require personalized attention. The result is a more efficient support process that enhances customer satisfaction and operational productivity.

Feedback for Continuous Improvement

To ensure that the support system remains effective and responsive to customer needs, a robust feedback mechanism is integrated into the system. This mechanism actively collects insights from customer interactions and support experiences, allowing the bank to identify trends, pain points, and areas for improvement. The collected feedback is analyzed to inform service enhancements and adapt the automated assistance features accordingly. This iterative process ensures that the support system evolves over time, leading to more effective solutions and a continuously improved customer experience.

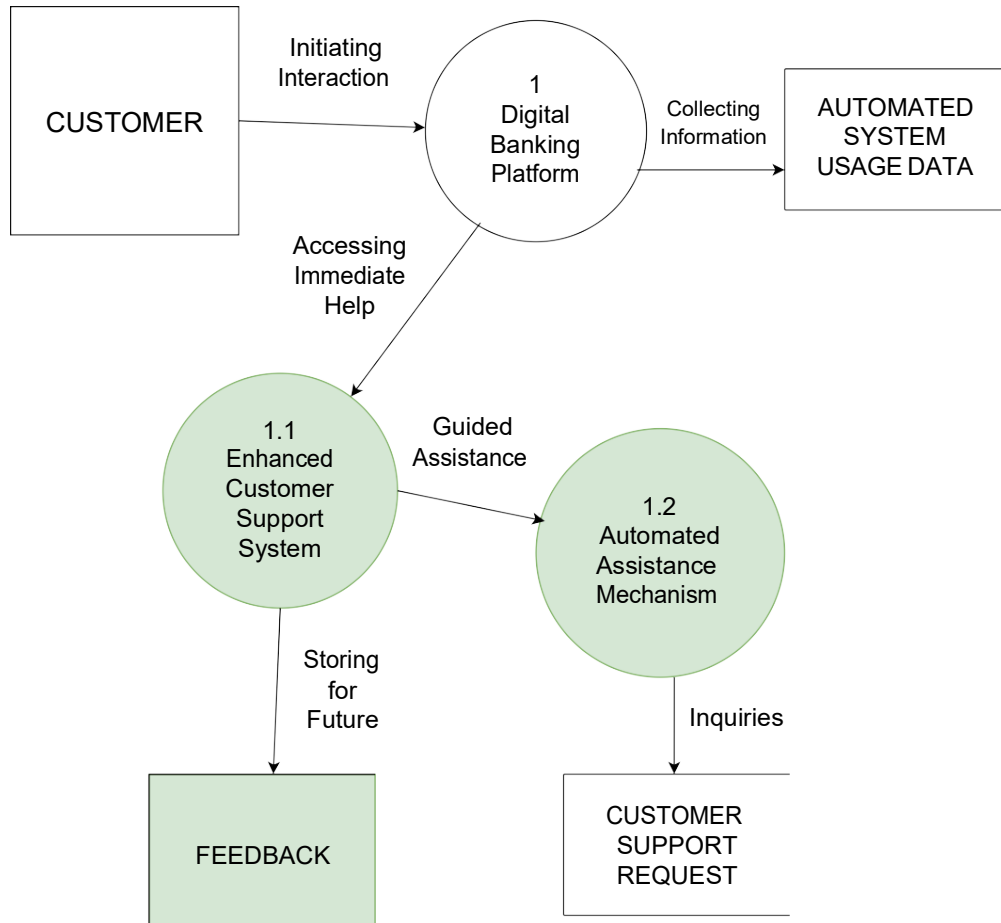


Figure 4.4: Proposed DFD for Automation Solutions to Mitigate Staff Shortages.(Changes are highlighted in green.)

4.2.5 Summary of Key Changes

The proposed digital banking system incorporates several key enhancements across multiple areas:

- **Improved Customer Engagement:** Personalized outreach through analytics, a loyalty program, and real-time support.
- **Automation and Proactive Solutions:** AI-driven support, real-time communication, and error resolution.
- **Enhanced Loan Approval Process:** Automated verification and risk management, reducing fraud and improving loan reliability.
- **Operational Efficiency:** Reducing human dependency through automation, leading to faster issue resolution and more efficient workflows.

These changes will lead to a more reliable, transparent, and efficient banking experience for IFIC Bank PLC's customers.

4.3 Database Design

Database design is the process of structuring data to efficiently store, manage, and retrieve information in a database system. It involves defining tables, fields, relationships, and constraints to ensure data integrity, optimize performance, and support the specific needs of an application or organization. A well-designed database not only organizes data logically but also enhances scalability, security, and ease of use. Key principles of database design include normalization, establishing relationships between entities, and adhering to best practices that ensure long-term maintainability and adaptability to evolving data requirements.

4.3.1 ER Diagram

Entity-Relationship (E-R) diagram design is a graphical representation technique used to model the entities, attributes, and relationships within a system or database. It helps in understanding the structure and organization of data, as well as the relationships between different entities. Here's an overview of the E-R diagram design process:

- **Identify Entities:** This step involves recognizing the key objects or concepts within the

system that need to store data.

- **Define Attributes:** In this phase, the properties of each entity that describe relevant details are determined.
- **Establish Relationships:** Here, the connections between entities are identified and defined.
- **Cardinality and Participation:** This involves specifying the number of instances involved in a relationship and the participation constraints.
- **Primary Keys and Foreign Keys:** This step includes choosing primary keys to uniquely identify records and foreign keys to link tables together.
- **Refine and Normalize:** At this stage, normalization techniques are applied to remove redundancy and improve data integrity.
- **Diagram Notation:** Entity-Relationship diagrams (ERDs) are utilized to visually represent entities, attributes, and relationships.
- **Review and Validation:** Finally, the design is validated by reviewing it for completeness, efficiency, and consistency with system requirements.

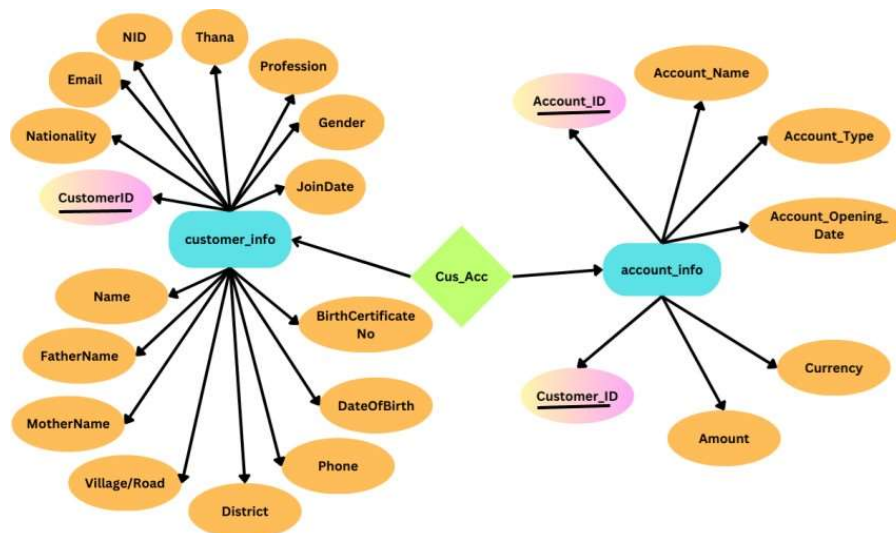


Figure 4.5: ER diagram for the customer info-account info relationship.

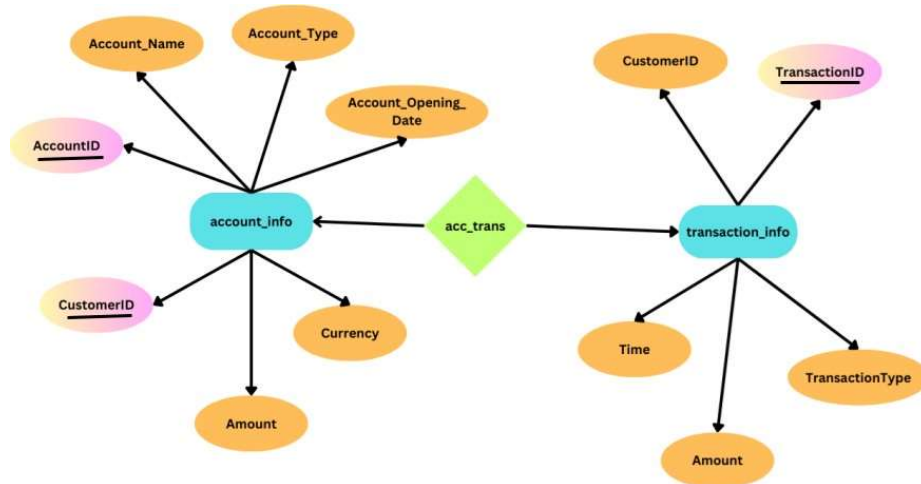


Figure 4.6: ER diagram for the account info-transaction info relationship.

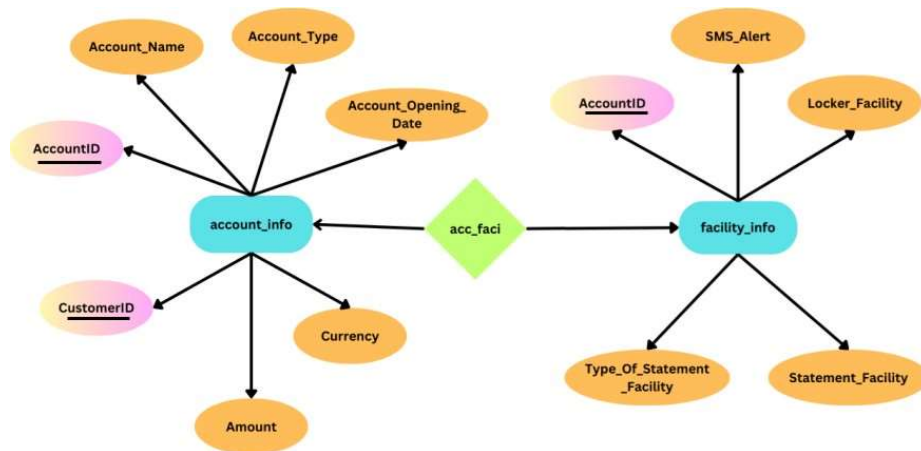


Figure 4.7: ER diagram for the account info-facility info relationship.

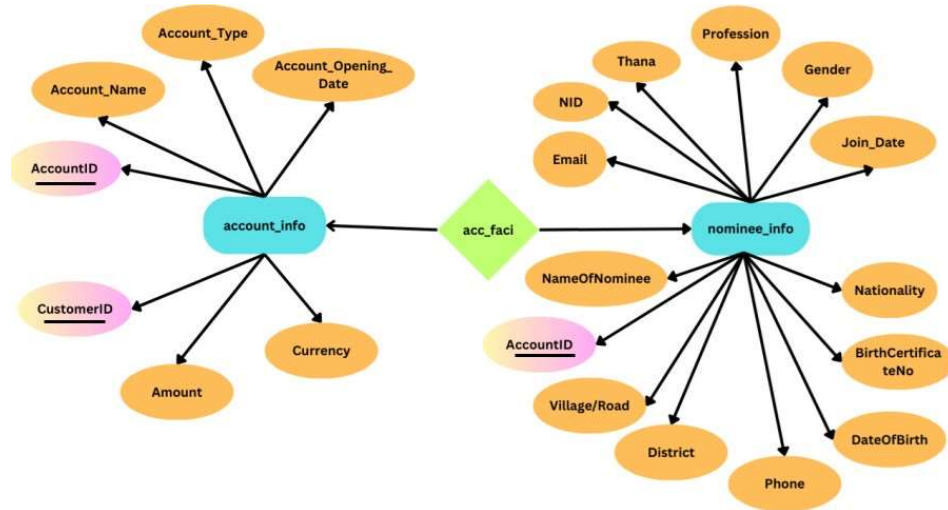


Figure 4.8: ER diagram for the account_info-nominee_info relationship.

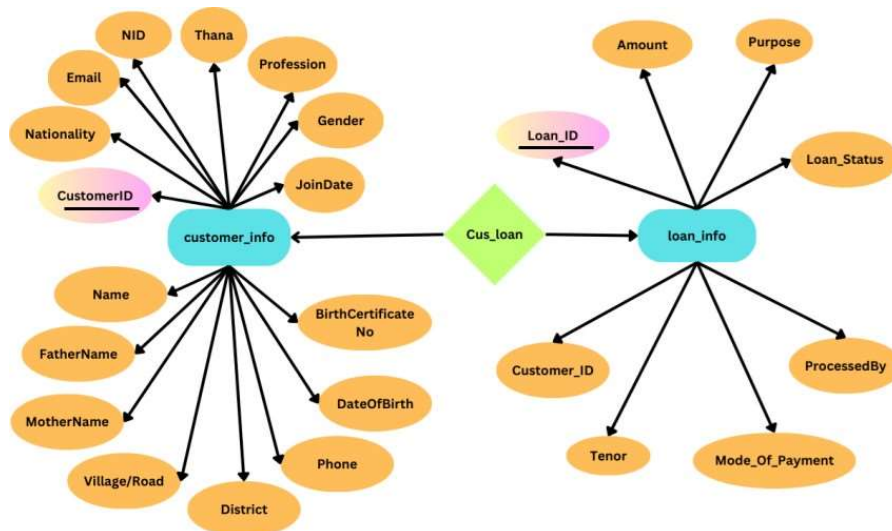


Figure 4.9: ER diagram for the customer_info-loan application relationship.

4.3.2 Database Relationships

Relation Between Customer Info and Account Info (One-to-Many)

A customer can open multiple accounts with the bank, such as savings, current, or fixed deposit accounts. This creates a one-to-many relationship where each customer (represented by their CustomerID in the customer info table) can have multiple accounts (linked via the CustomerID in the account info table). This structure allows the bank to maintain comprehensive records for each account a customer owns.

This relationship benefits both the bank and the customer by facilitating better account management. From a customer perspective, having multiple accounts linked to a single identifier streamlines access to banking services, while allowing for flexibility in account types. For the bank, it ensures efficient record-keeping and comprehensive customer profiles, which are essential for personalized services like cross-selling or up-selling financial products.

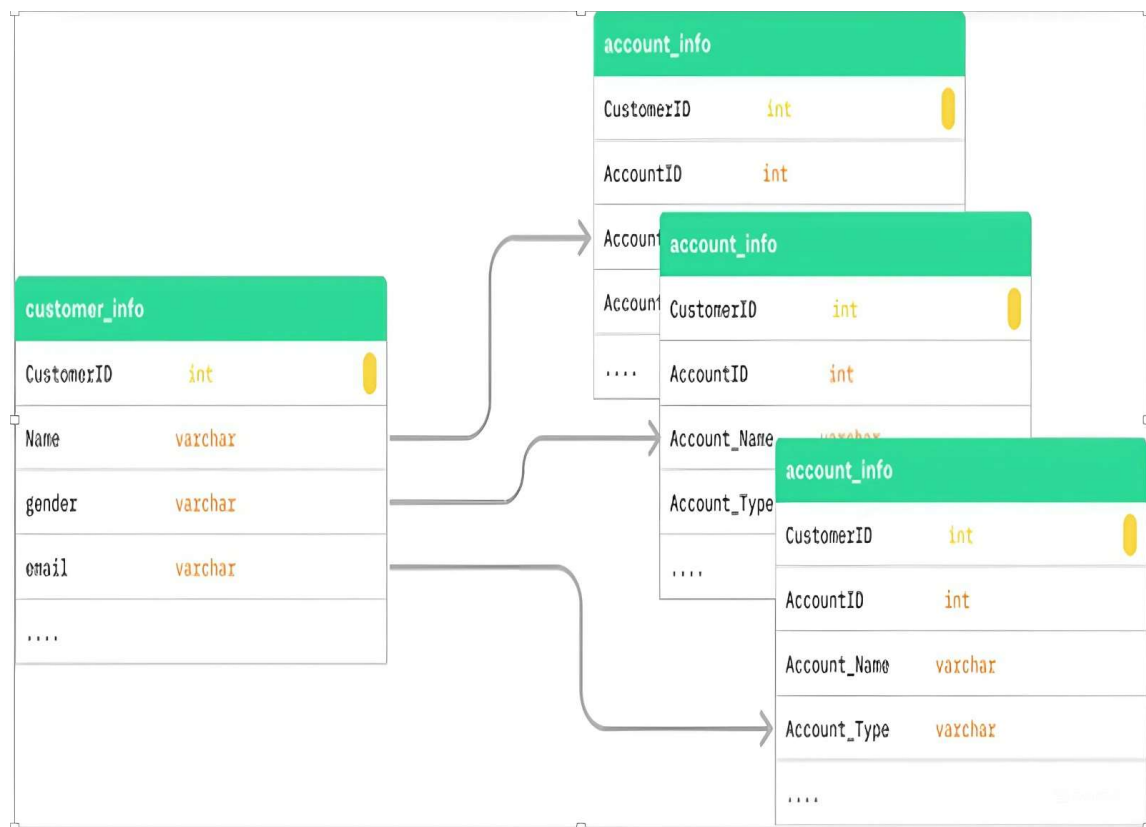


Figure 4.10: Relation Between customer.info and account.info table (one-to-many).

Relation Between Account Info and Facility Info (One-to-One)

Each account is eligible for specific banking facilities such as SMS alerts, locker services, and account statements. This relationship is one-to-one, meaning each account is linked to one specific set of facilities. This ensures that every account has unique facility settings tailored to the customer’s preferences or needs.

Additionally, the one-to-one relationship ensures that facility management is both simplified and secure. It enables precise customization of services for each account, which improves the user experience by catering to individual customer requirements. This structure also reduces the chances of service overlap or misallocation of facilities, ensuring that each customer’s banking experience is streamlined, personalized, and secure.

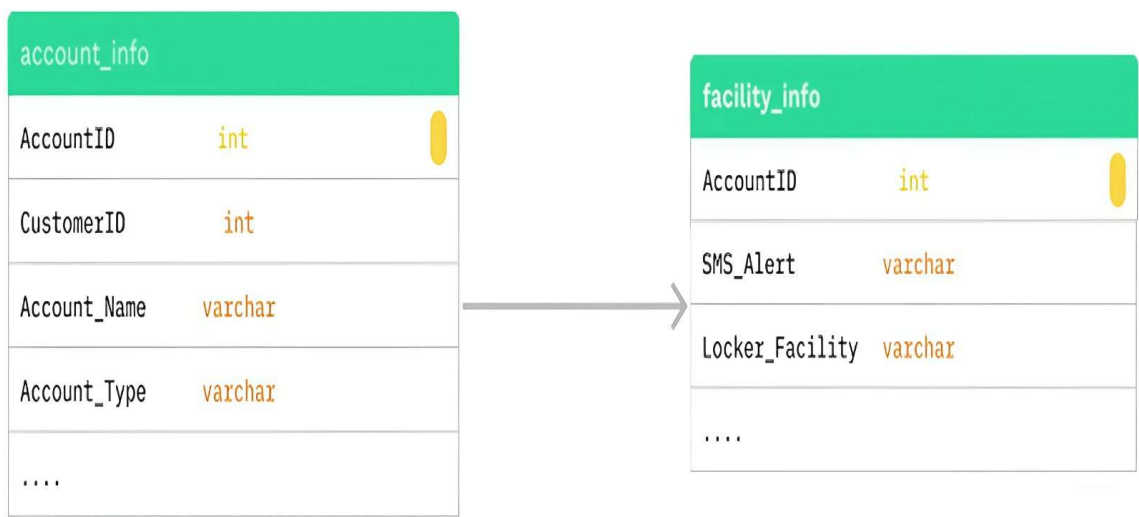


Figure 4.11: Relation Between account_info and facility info table (one-to-one).

Relation Between Account Info and Nominee Info (One-to-One)

For every account, a customer can nominate a person (usually a family member) to be the nominee. This nominee will inherit or receive benefits from the account in case of the account holder’s death. The one-to-one relationship ensures that each account can only have one nominee associated with it at any given time. This prevents confusion and ensures that the nominee details are clearly tied to each account.

Furthermore, this structure enhances clarity and legal certainty in the event of inheritance or claim disputes. By limiting each account to a single nominee, the bank simplifies the process of transferring account ownership, reducing administrative delays. This arrangement also ensures that nominee updates are seamlessly integrated, making it easier for customers to change their designated nominee if necessary, thus fostering better customer satisfaction and legal compliance.

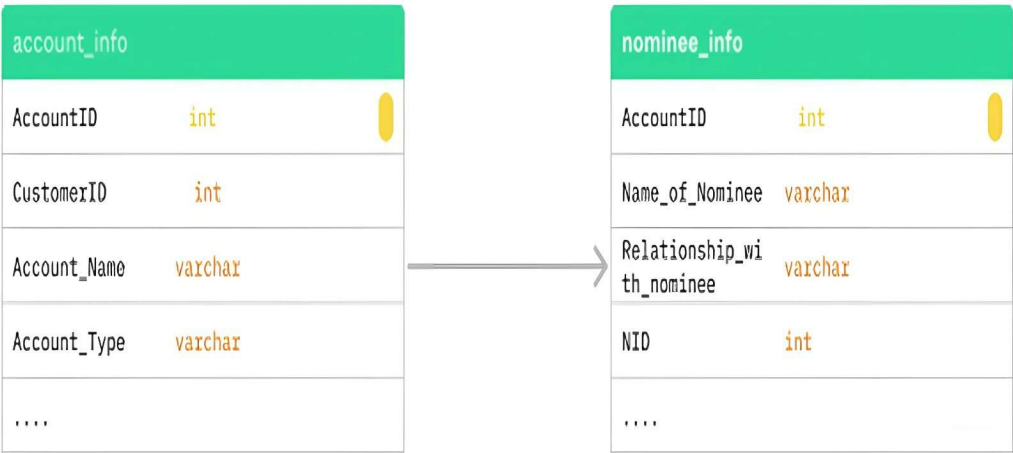


Figure 4.12: Relation Between account info and nominee.info table (one-to-one).

Relation Between Customer Info and Transaction Info (One-to-Many)

Each customer may conduct numerous transactions over time, such as deposits, withdrawals, and transfers. These transactions are recorded in the transaction info table, with each transaction linked back to the respective customer through the CustomerID. This one-to-many relationship means that one customer can have multiple transaction records, providing a detailed transaction history for auditing and tracking purposes.

Additionally, this structure ensures accurate financial tracking for both customers and the bank, aiding in identifying spending patterns, fraud detection, and regulatory compliance. By maintaining a comprehensive transaction history, the bank can also offer personalized financial insights and recommendations to its customers, enhancing the overall banking experience. This relationship enables smoother reconciliation processes and facilitates quicker resolution of disputes, as every transaction is meticulously tied to a single customer.

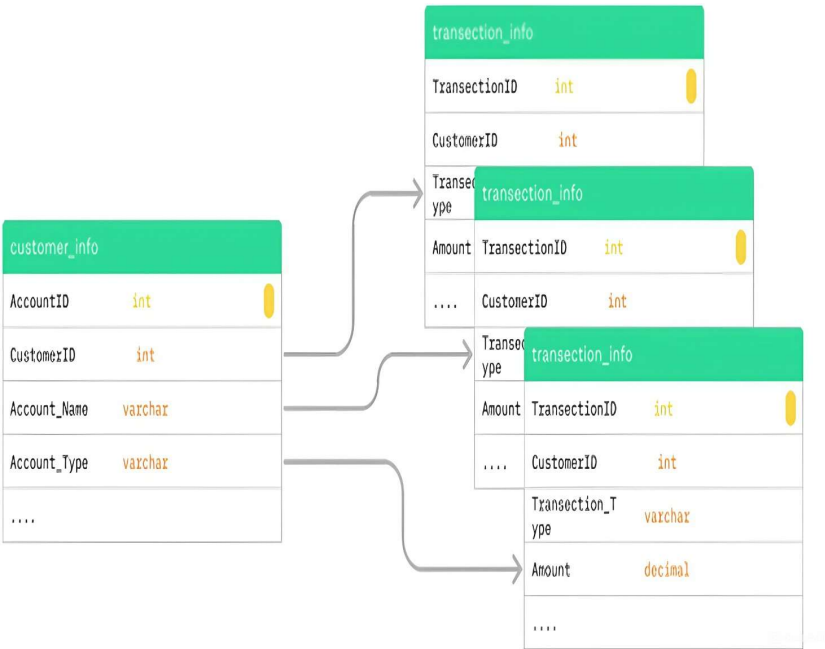


Figure 4.13: Relation Between customer_info and transaction_info table(one-to-many).

Relation Between Customer Info and Loan Application (One-to-Many)

A customer may apply for multiple loans over time, such as home loans, personal loans, or car loans. Each loan application is recorded in the loan application table, and the loan records are linked to the customer using the CustomerID. This one-to-many relationship allows for multiple loan applications to be tied to a single customer, enabling the bank to manage each loan’s details, such as amount, purpose, and status.

This structure enhances the bank’s ability to track the financial obligations of a customer across different loans, facilitating more accurate risk assessment and loan management. By linking each loan application to the same customer, the bank can provide a holistic view of a customer’s borrowing history, aiding in decision-making processes like offering future loans or renegotiating terms. Additionally, the system supports customized loan products and better loan servicing, as it allows for comprehensive tracking of repayment behaviors and overall financial stability, ensuring a more tailored banking experience for customers.

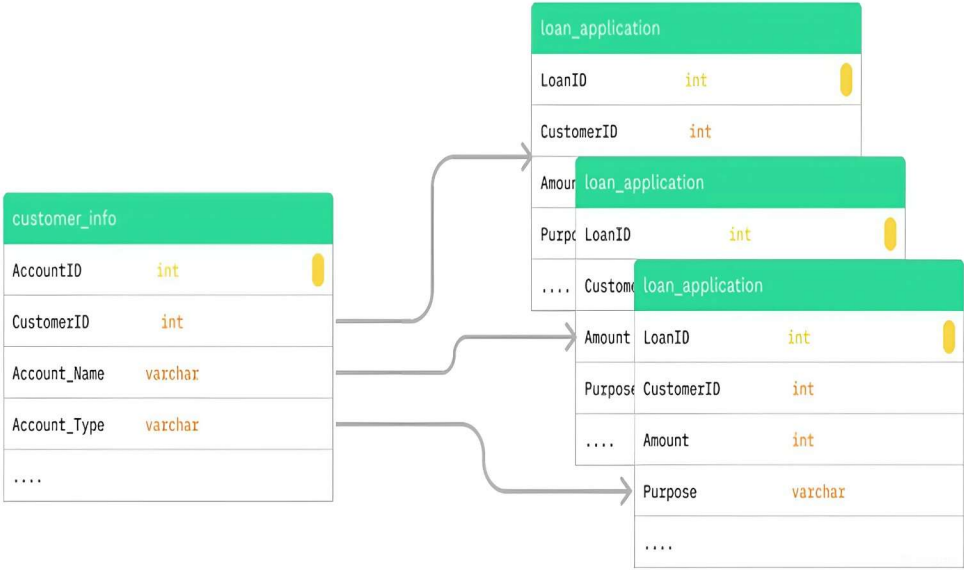


Figure 4.14: Relation Between customer_info and loan_application table(one- to-many).

This relational structure ensures data consistency and integrity across different aspects of the banking system, enabling efficient tracking and management of customer data, accounts, transactions, and loan applications.

4.3.3 Database Tables (Data Type and Size)

A database table is a collection of related data organized in rows and columns. Tables are fundamental components of a relational database and serve as the main structure for storing and organizing data. Our goal is to develop a database that encompasses all customer information, including personal details, nominee details, transaction details, and other relevant data.

Customer Information Table (`customer_info`)

The `customer_info` table includes essential customer details, such as: **CustomerID, Name, Father's Name, Mother's Name, Gender, Date of Birth, NID number, Birth Certificate number, Profession, Nationality, Phone number, Email Address, Address (Village or Road), Thana, District, Join Date.**

Account Information Table (`account_info`)

The `account_info` table records details of customer accounts, including: **AccountID, CustomerID, Account Name, Account Type, Amount, Currency, Account Opening Date.**

Facility Information Table (`facility_info`)

The `facility_info` table encompasses the following fields: **AccountID, SMS Alert, Locker Facility, Statement Facility, Type of Statement Facility.**

Nominee Information Table (`nominee_info`)

The `nominee_info` table details nominees for accounts, containing: **AccountID, CustomerID, Name of Nominee, Relationship with Applicant, NID number, Birth Certificate number, Date of Birth, Phone number, Email address, Address (Village or Road), Thana, District, Country.**

Transaction Information Table (`transaction_info`)

The `transaction_info` table captures transaction details, which include: **TransactionID, CustomerID, Transaction Type, Amount, Time.**

Loan Application Table (loan application)

The `loan_application` table records loan details, featuring: **LoanID, CustomerID, Amount, Purpose, Tenor, Mode of Payment, Loan Status, Processed By.**

However, storing all this information in a single database table would result in redundancy, leading to significant storage inefficiencies for the bank. Therefore, to ensure cost efficiency, we will implement database normalization techniques. Additionally, using a single column as a primary key will not suffice for unique identification of each entry. Instead, we will employ composite keys, which might increase the complexity of our table structure.

By employing normalization techniques, we will decompose the redundant table and create several separate tables with related data, thereby eliminating redundancy from the database. Each table will have its own primary key, and we will establish the necessary relationships between them.

To achieve this, we will create a relational database consisting of tables such as `customer_info`, `account_info`, `nominee_info`, `facility_info`, and `transaction_info`.

In the `account_info` table, `AccountID` is made the primary key while `CustomerID` is the foreign key, which means that a customer may have different accounts in the same bank. In the `facility_info` table, `AccountID` is made both the primary and foreign key. The same applies to the `nominee_info` table, where `AccountID` is marked as both primary and foreign key. In the `transaction_info` table, `TransactionID` is made the primary key, while `CustomerID` is designated as the foreign key, as one customer might have many transactions in the bank. Lastly, the `loan_application` table includes `CustomerID` as its foreign key, which establishes a one-to-many relationship.

#	Name	Type
1	CustomerID 	int(11)
2	Name	varchar(100)
3	FatherName	varchar(100)
4	MotherName	varchar(100)
5	Gender	varchar(10)
6	DateOfBirth	date
7	NID	int(11)
8	BirthCertificateNo	int(11)
9	Profession	varchar(100)
10	Nationality	varchar(100)
11	Phone	varchar(20)
12	Email	varchar(100)
13	VillageOrRoad	varchar(100)
14	Thana	varchar(100)
15	District	varchar(100)
16	JoinDate	date

Figure 4.15: Field Name and Data types of customer.info Table.

#	Name	Type
1	AccountID 	int(11)
2	CustomerID  	int(11)
3	Account_Name	varchar(100)
4	Account_Type	varchar(100)
5	Amount	decimal(12,2)
6	Currency	varchar(10)
7	Account_Opening_Date	date

Figure 4.16: Field Name and Data types of account.info Table.

#	Name	Type
1	AccountID 	int(11)
2	SMS_Alert	varchar(10)
3	Locker_Facility	varchar(10)
4	Statement_Facility	varchar(10)
5	Types_Of_Statement_Facility	varchar(50)

Figure 4.17: Field Name and Data types of facility.info Table.

#	Name	Type
1	AccountID 	int(11)
2	Name_of_Nominee	varchar(50)
3	Relationship_with_applicant	varchar(50)
4	NID	varchar(20)
5	Birth_Certificate_No	varchar(20)
6	Date_of_Birth	date
7	Phone	varchar(20)
8	Email	varchar(50)
9	Village_Road	varchar(100)
10	Thana	varchar(50)
11	District	varchar(50)
12	Country	varchar(50)

Figure 4.18: Data types of nominee_info attributes.

#	Name	Type
1	LoanID 	int(11)
2	CustomerID 	int(11)
3	Amount	decimal(10,2)
4	Purpose	varchar(255)
5	Tenor	int(11)
6	ModeOfPayment	varchar(50)
7	ProcessedBy 	int(11)
8	Loan_Status	int(11)

Figure 4.19: Data type of loan_application attributes.

#	Name	Type
1	TransactionID 	int(11)
2	CustomerID 	int(11)
3	Transaction_Type	varchar(100)
4	Amount	decimal(12,2)
5	Time	datetime

Figure 4.20: Data type of transaction_info attributes.

4.4 Form Design

Forms of any organization represent what kind of organization that is, what it will do as well as their goals. A form is used to collect data. Without data, there is no system. In many cases, to get this precise data to make an impact on our system, we need to have data forms to collect and capture information directly. This helps automate data collection, thus saving time and creating more accurate reports. Forms elicit and capture information required by organizational members that will often be input to the computer. Every form represents individual things. To update and requirements of the form some things are explained in this part.

Modernized Account Opening Form

This redesign focuses on creating an interactive and user-friendly form that includes essential fields, real-time validation, and a simplified user interface. The form will also offer various digital enhancements for a better overall experience.

Digital Form Enhancements

- **Section 1: Personal Information**
 - **NID/Smart ID Validation:** Automatically validate against a central database.
 - **Date of Birth:** Integrate a calendar tool for easier selection.
 - **Name (in English/Bangla):** Auto-populate the name in Bangla based on the English input. This can also be pre-filled using NID/Smart ID data.
 - **Customer Short Name:** Optional field with a real-time character limit.
 - **Father's & Mother's Name:** Auto-complete using NID/Smart ID data.
- **Section 2: Marital Status & Spouse Details**
 - **Marital Status:** Dropdown menu with predefined options (Single, Married, etc.).
 - **Spouse Name:** Auto-complete functionality.
- **Section 3: Address Details**
 - **Geolocation Integration:** Use Google Maps API to allow users to pinpoint their address, reducing input errors.
 - **Address Fields:** Quick-fill options for cities and postcodes using lookup tools.
- **Section 4: Contact Information**
 - **Primary Contact Number:** Real-time validation with SMS-based OTP.
 - **Alternate Contact Number:** Optional input.
 - **Email Address:** Include an option to receive monthly statements via email.
- **Section 5: Employment & Additional Information**

- **Source of Funds:** Dropdown categories such as Salary, Business Income, Savings, etc.
- **Profession:** Dropdown options with broad categories and relevant subcategories.
- **Section 6: Nationality & Residency**
 - **Dual Nationality Option:** A field to specify secondary citizenship, if applicable.
- **Section 7: Additional Features**
 - **SMS and Mobile Banking Registration:** A checkbox allowing users to opt-in for these services.
 - **Document Upload:** Easily upload scanned copies of national ID, address proof, and a recent photo, streamlining the process.

The modernized Account Opening Form for IFIC Bank PLC will significantly enhance user experience by incorporating real-time validation, intuitive dropdowns, and automated data entry features. The integration of tools like Google Maps API, SMS-based OTP validation, and easy document uploads will streamline the process, reduce errors, and improve overall efficiency. These updates aim to simplify the user journey while maintaining the highest standards of accuracy and security.

The following section presents images of the newly designed forms, showcasing their enhanced features and functionality. These new forms have been meticulously crafted to address the specific needs of users while ensuring a more streamlined interaction process.

In this redesign, a focus was placed on improving usability and accessibility, allowing users to navigate through the forms with greater ease and efficiency. Each stage incorporates intuitive layouts and clearly labeled fields, which reduce confusion and minimize the time required to complete them. Additionally, enhancements such as auto-fill capabilities, real-time validation, and responsive design elements have been integrated to further elevate the user experience.

Personal Account Application Form

Personal Information

NID/Smart ID:

Date of Birth (D/M/Y):



Name (in English):

Name (in Bangla):

Customer Short Name (in English):

Father's Name (in English):

Mother's Name (in English):

Nominee Information

NID/Smart ID:

Date of Birth (D/M/Y):



Figure 4.21: Simplistic Overview of the New Form.

Name (in English):

Name (in Bangla):

Relation with Nominee:

Marital Status & Spouse Details

Marital Status:

Spouse Name (in English):

Permanent & Present Address

Permanent Address:

Division:

District:

Police Station:

Post Code:

Figure 4.22: Auto-Fill and Validation Features as Mentioned for the New Form.

Present Address:

Monnafer Mor Bir Sreshtho Shaheed Captain Mohiuddin Jahangir Smarani,
Rajshahi 6204

Pin Your Address



Contact Information

Primary Contact Number:

Alternate Contact Number:

Email Address:

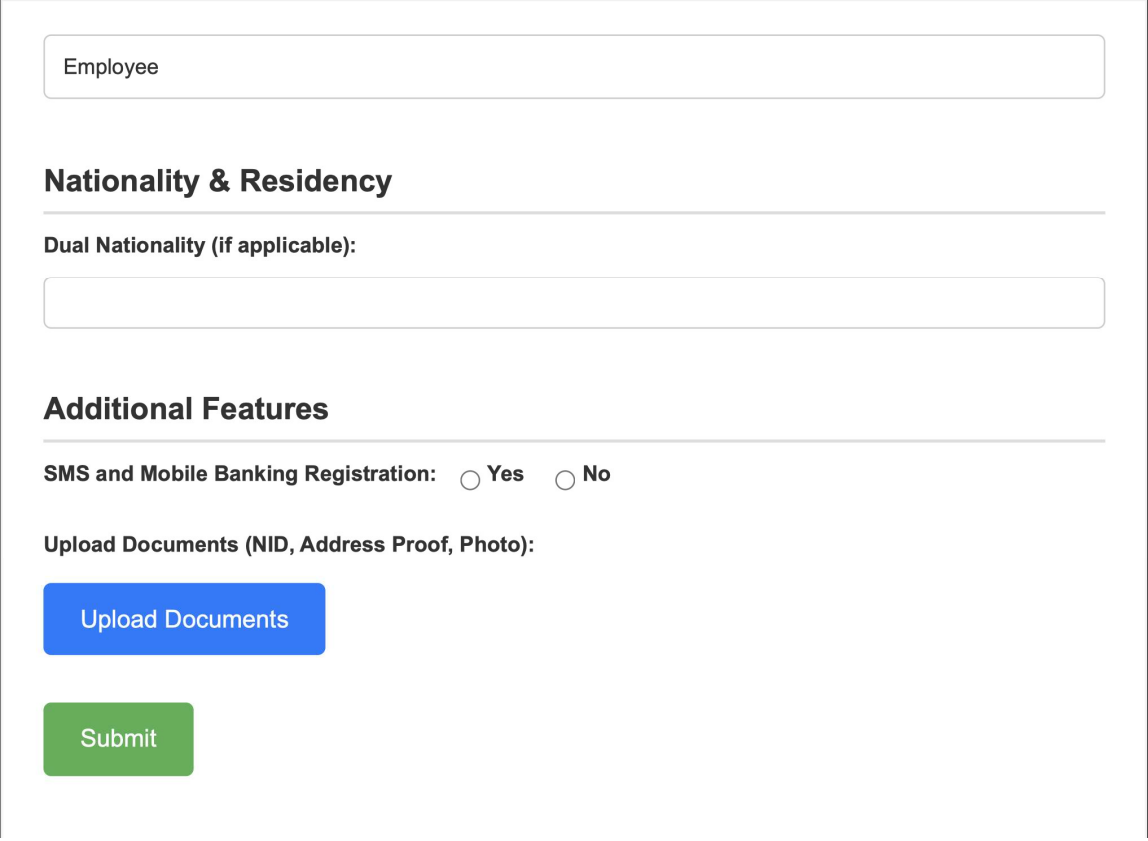
Employment & Additional Information

Source of Funds:

Salary

Profession:

Figure 4.23: Data Fetching from Real-time Location on the new Form.



The image shows a web form interface for submission. At the top, there is a text input field containing the word "Employee". Below this is a section header "Nationality & Residency" followed by a horizontal line. Underneath is the label "Dual Nationality (if applicable):" and an empty text input field. The next section is "Additional Features", also followed by a horizontal line. Below this, there is a label "SMS and Mobile Banking Registration:" followed by two radio buttons, one labeled "Yes" and one labeled "No". Underneath is the label "Upload Documents (NID, Address Proof, Photo):" and a blue button labeled "Upload Documents". At the bottom of the form is a green button labeled "Submit".

Figure 4.24: Final Submission Interface of the New Form.

Collecting the data needed from the users can be critical for a system and forms are really the way to make that happen. Hence, we tried to design the form carefully so that important data related to the system can be collected and processed correctly. We also followed the guidelines for making the form useful to the user by making the forms easy to fill in, ensuring that forms meet the purpose for which they are designed and by keeping the form attractive.

4.5 Conclusion

Systems design is a vital phase in the development of any system, as it translates the requirements from systems analysis into a structured and functional model. The careful design of input forms and output reports plays a key role in ensuring the system is user-friendly and efficient. Forms serve as essential tools for collecting accurate data, while databases organize this data for easy access and efficient processing. Together, these elements form the backbone

of a well-designed system that meets organizational goals and facilitates smooth operations. By integrating both forms and databases effectively, systems design ensures a system that not only performs reliably but also supports informed decision-making across the organization.